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Medical Image Segmentation in a Multiple Labelers Context: Application to the Study of Histopathology

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Segmentación de Imágenes Médicas en un Contexto de Múltiples Anotadores: Aplicación al Estudio de Histopatologías

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56 meaningfully to the field of early cancer detection.

ABSTRACT

Breast cancer remains one of the deadliest diseases affecting women worldwide, making accurate diagnosis and treatment planning crucial for improving patient outcomes. Semantic segmentation of histopathology images plays a vital role in this process, as it enables precise identification and analysis of tissue structures. However, the current approaches face significant challenges in real-world scenarios where multiple annotators with varying expertise and reliability contribute to the labeling process. This crowdsourcing paradigm, while cost-effective, introduces inconsistencies that can compromise the quality of the segmentation results.

This thesis addresses two fundamental challenges in medical image segmentation: the inherent uncertainty in multiple annotator scenarios and the need for robust segmentation models that can handle varying image quality and annotator expertise. These challenges arise from the complex nature of histopathology images, where tissue structures can be ambiguous, and the varying levels of expertise among annotators lead to inconsistent labeling patterns. Additionally, the high cost and time requirements for expert annotations create a need for efficient solutions that can work with crowdsourced data.

The first objective introduces a novel loss function that implicitly infers optimal segmentation while assessing labeler reliability without requiring explicit

performance inputs. This approach differs from conventional methods by eliminating the need for direct supervision of labeler performance, instead using an intermediate reliability map that enables the model to prioritize information from reliable labelers while disregarding noisy labels. The mathematical foundation of this approach lies in the probabilistic modeling of annotator reliability across the input space.

The second objective focuses on the development of a robust segmentation architecture that combines the strengths of U-shaped deep learning models with the proposed reliability-aware loss function. This combination enables the model to learn both the segmentation task and the reliability patterns simultaneously, leading to more accurate and consistent results. The architecture's key innovation lies in its ability to adapt to varying image quality and annotator expertise levels without requiring explicit performance metrics.

The third objective validates the proposed approach through extensive experimentation on both synthetic datasets and real-world histopathology images. The evaluation demonstrates superior performance compared to state-of-the-art approaches in terms of segmentation accuracy and uncertainty quantification. The results show that the proposed method can effectively handle inconsistent annotations while maintaining high segmentation accuracy, making it particularly valuable for clinical applications.

The research makes significant contributions to the field by providing a robust solution for handling inconsistent annotations in medical image segmentation, particularly valuable in histopathology where expert annotations are expensive and time-consuming to obtain. The proposed approach shows promising results in reducing annotation costs while maintaining high segmentation accuracy, making it particularly relevant for clinical applications where precise segmentation of tissue structures is crucial for diagnosis and treatment planning. This work opens new research directions in the areas of crowdsourced medical image analysis, uncertainty quantification in deep learning, and the development of more efficient annotation protocols for medical imaging.

109 **Keywords:** Crowdsourcing, Multiple Annotators, Histopathology, Breast Cancer,
110 Semantic Image Segmentation, Gaussian Processes, Deep Learning, Medical Image
111 Analysis

113 El cáncer de mama sigue siendo una de las enfermedades más mortales en la
114 población femenina en todo el mundo, lo que hace que el diagnóstico preciso y la
115 planificación del tratamiento sean cruciales para mejorar las condiciones de los
116 pacientes. La segmentación semántica de imágenes histopatológicas juega un
117 papel vital en este proceso, ya que permite la identificación y análisis preciso de
118 estructuras tisulares. Sin embargo, los enfoques actuales enfrentan desafíos
119 significativos en escenarios del mundo real donde múltiples anotadores con
120 diferentes niveles de experiencia y confiabilidad contribuyen al proceso de
121 etiquetado. Este paradigma de crowdsourcing, aunque rentable, introduce
122 inconsistencias que pueden comprometer la calidad de los resultados de
123 segmentación.

124 Esta tesis aborda dos desafíos fundamentales en la segmentación de imágenes
125 médicas: la incertidumbre inherente en escenarios de múltiples anotadores y la
126 necesidad de modelos de segmentación robustos que puedan manejar la calidad
127 variable de las imágenes y la experiencia de los anotadores. Estos desafíos surgen
128 de la naturaleza compleja de las imágenes histopatológicas, donde las estructuras
129 tisulares pueden ser ambiguas, y los diferentes niveles de experiencia entre los
130 anotadores conducen a patrones de etiquetado inconsistentes. Además, el alto
131 costo y los requisitos de tiempo para las anotaciones de expertos crean la
132 necesidad de soluciones eficientes que puedan trabajar con datos crowdsourced.

133 El primer objetivo introduce una función de pérdida novedosa que infiere
134 implícitamente la segmentación óptima mientras evalúa la confiabilidad de los
135 etiquetadores sin requerir entradas explícitas sobre su rendimiento. Este enfoque
136 difiere de los métodos convencionales al eliminar la necesidad de supervisión
137 directa del rendimiento de los etiquetadores, utilizando en su lugar un mapa de
138 confiabilidad intermedio que permite al modelo priorizar la información de
139 etiquetadores confiables mientras descarta etiquetas ruidosas. La base matemática
140 de este enfoque radica en el modelado probabilístico de la confiabilidad de los
141 anotadores a través del espacio de entrada.

142 El segundo objetivo se centra en el desarrollo de una arquitectura de segmentación
143 robusta que combina las fortalezas de los modelos de aprendizaje profundo en
144 forma de U con la función de pérdida propuesta que tiene en cuenta la
145 confiabilidad. Esta combinación permite que el modelo aprenda tanto la tarea de
146 segmentación como los patrones de confiabilidad simultáneamente, lo que lleva a
147 resultados más precisos y consistentes. La innovación clave de la arquitectura
148 radica en su capacidad para adaptarse a diferentes niveles de calidad de imagen y
149 experiencia de los anotadores sin requerir métricas de rendimiento explícitas.

150 El tercer objetivo valida el enfoque propuesto a través de experimentación
151 exhaustiva tanto en conjuntos de datos sintéticos como en imágenes
152 histopatológicas del mundo real. La evaluación demuestra un rendimiento
153 superior en comparación con los enfoques más avanzados en términos de
154 precisión de segmentación y cuantificación de incertidumbre. Los resultados
155 muestran que el método propuesto puede manejar efectivamente anotaciones
156 inconsistentes mientras mantiene una alta precisión de segmentación, lo que lo
157 hace particularmente valioso para aplicaciones clínicas.

158 La investigación hace contribuciones significativas al campo al proporcionar una
159 solución robusta para manejar anotaciones inconsistentes en la segmentación de
160 imágenes médicas, particularmente valiosa en histopatología donde las
161 anotaciones de expertos son costosas y requieren mucho tiempo para obtenerlas.

162 El enfoque propuesto muestra resultados prometedores en la reducción de costos
163 de anotación mientras mantiene una alta precisión de segmentación, lo que lo hace
164 particularmente relevante para aplicaciones clínicas donde la segmentación precisa
165 de estructuras tisulares es crucial para el diagnóstico y la planificación del
166 tratamiento. Este trabajo abre nuevas direcciones de investigación en las áreas de
167 análisis de imágenes médicas crowdsourced, cuantificación de incertidumbre en
168 aprendizaje profundo y el desarrollo de protocolos de anotación más eficientes
169 para imágenes médicas.

170 **Palabras clave:** Crowdsourcing, Múltiples Anotadores, Histopatología, Cáncer de
171 Mama, Segmentación Semántica de Imágenes, Procesos Gaussianos, Aprendizaje
172 Profundo, Análisis de Imágenes Médicas

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352		truth annotation is not expressive enough, the Dice coefficient	
353		remains below 0.55, which is not considered a good performance.	
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ABBREVIATIONS

371	CAD	Computer-Aided Diagnosis 2, 5, 6
372	CCGP	Correlated Chained Gaussian Processes 17
373	CCGPMA	Correlated Chained Gaussian Processes for Multiple Annotators 17, 18, 51, 53,
374		54, 58, 83
375	CE	Cross Entropy 60, 63
376	CGP	Chained Gaussian Processes 17
377	CKA	Centered Kernel Alignment 42
378	CNN	Convolutional Neural Networks 3, 14, 23, 24, 65, 83
379	CT	Computed Tomography 1, 2, 11
380	ELBO	Evidence Lower Bound 18
381	GCE	Generalized Cross Entropy 60, 63
382	GCECDL	Generalized Cross-Entropy-based Chained Deep Learning 20
383	ISS	Image Semantic Segmentation 2, 3, 6, 9, 13, 20, 22–24, 36, 81, 83
384	KAAR	Kernel Alignment-Based Annotator Relevance Analysis 41
385	LF	Latent Function 17
386	MAE	Mean Absolute Error 60, 63
387	MITs	Medical Imaging Techniques 1
388	ML	Machine Learning 9, 11, 20
389	MRI	Magnetic Resonance Imaging 1, 2
390	MV	Majority Voting 9, 11
391	OCR	Optical Character Recognition 11
392	PET	Positron Emission Tomography 13
393	ROI	Region of Interest 2, 6, 15, 31
394	SLFM	Semi-Parametric Latent Factor Model 17, 49, 51
395	SS	Semantic segmentation 3
396	STAPLE	Simultaneous Truth and Performance Level Estimation 12–14, 36
397	WSI	Whole Slide Imaging 1, 5, 6, 8, 15, 29, 39

398

399

CHAPTER

400

ONE

401

402

INTRODUCTION

403

1.1 Motivation

404 Since Roentgen's discovery of X-rays in 1895, medical imaging has advanced
405 significantly, with modalities like radionuclide imaging, ultrasound, **Computed**
406 **Tomography (CT)**, **Magnetic Resonance Imaging (MRI)**, and digital radiography
407 emerging over the past 50 years. Modern imaging extends beyond image
408 production to include processing, display, storage, transmission and analysis.
409 [Zhou et al., 2021]. Other **Medical Imaging Techniques (MITs)** have arose during
410 the last decades, some of them implying only the examination of certain pieces or
411 tissues instead of complete patients, like histopathological images, which are
412 images of tissue samples obtained from biopsies or surgical resections and are
413 widely used for the diagnosis of diseases like cancer through **Whole Slide Imaging**
414 (**WSI**) scanners [Rashmi et al., 2021].

415 Along with the advances in technologies for medical images acquisition,
416 computational technologies on pattern recognition and artificial intelligence have

also emerged, allowing the development of **Computer-Aided Diagnosis (CAD)** systems based on machine learning algorithms. These systems aim to assist physicians in the diagnosis and treatment of diseases, by providing a second opinion or by automating the analysis of medical images. [Panayides et al., 2020]. One of the most used tasks in which machine learning technologies is being used in the universe of medical images is **Image Semantic Segmentation (ISS)**, which consists of assigning a label to each pixel in an image according to the object it belongs to. This task is crucial for the development of **CAD** systems, as it allows the identification of **Region of Interest (ROI)** in the images, which can be used to detect and classify diseases [Azad et al., 2024].

The application of Machine Learning in medical imaging has grown significantly, with key tasks including classification, segmentation, anomaly detection, super-resolution, image registration, and synthetic image generation [Brito-Pacheco et al., 2025]. Among imaging modalities, X-rays and **CT** scans are widely used for classification and anomaly detection, especially in pulmonary and oncological applications. **MRI** and ultrasound play a crucial role in segmentation and resolution enhancement, while PET/SPECT imaging is essential for anomaly detection in oncology and neurodegenerative diseases [Brito-Pacheco et al., 2025]. Histopathology is rapidly gaining prominence, particularly in segmentation and feature extraction, where AI-driven techniques aid in automated cancer diagnosis and tissue structure analysis. The integration of Deep Learning in histological image processing is revolutionizing pathology, enabling more precise and efficient diagnostics. A brief comparison of the tasks and medical image types based on recent literature review, can be seen in Figure 1-1. [Yu et al., 2025], [Brito-Pacheco et al., 2025], [Ryou et al., 2025], [Hu et al., 2025], [Elhaminia et al., 2025]

For solving the different requirements of tasks in medical images, a variety of computational techniques have been developed [Zhou et al., 2021]. Initially, these needs were covered with simple morphological filters, which implied no training process or elaborated optimization. However, as the complexity of the tasks increased, the need for more sophisticated techniques arose, leading to the

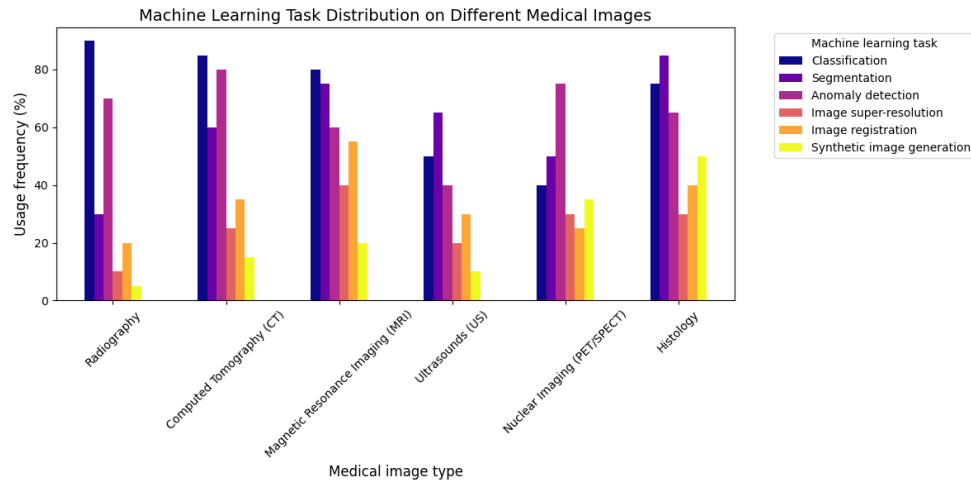


Figure 1-1 Estimation of the popularity of tasks and medical image types based on recent literature review (count of referenced terms).

application of advanced statistical tools and machine learning algorithms like Support Vector Machines, Decision Trees, and SGD Neural Networks [Avanzo et al., 2024]. The coevolution of advances in medical image acquisition, computational power (i.e. Moore's law) and statistical/mathematical techniques have led to a convergence for merging state of the art algorithms with medical imaging [Shalf, 2020]. Figure 1-2 shows a brief timeline of coevolution between some conspicuous advances in computational pattern recognition and its medical applications in different scopes (besides medical imaging) [Avanzo et al., 2024].

Convolutional Neural Networks (CNN) have been widely used in Semantic segmentation (SS) tasks, as they have outperformed traditional machine learning algorithms in this task for both medical and non medical images [Xu et al., 2024] [Sarvamangala and Kulkarni, 2022]. However, most CNN architectures are deep, which imply a necessity of a large amount of data to train them. This introduces a problem since both the acquisition and annotation of medical images are expensive and time-consuming processes. This is especially true for ISS tasks, as they require pixel-level annotations, which is taxing in terms of cost, time and logistics involved [Bhalgat et al., 2018]. Other fashions face this problem through less expensive annotation strategies like bounding boxes or anatomical landmarks

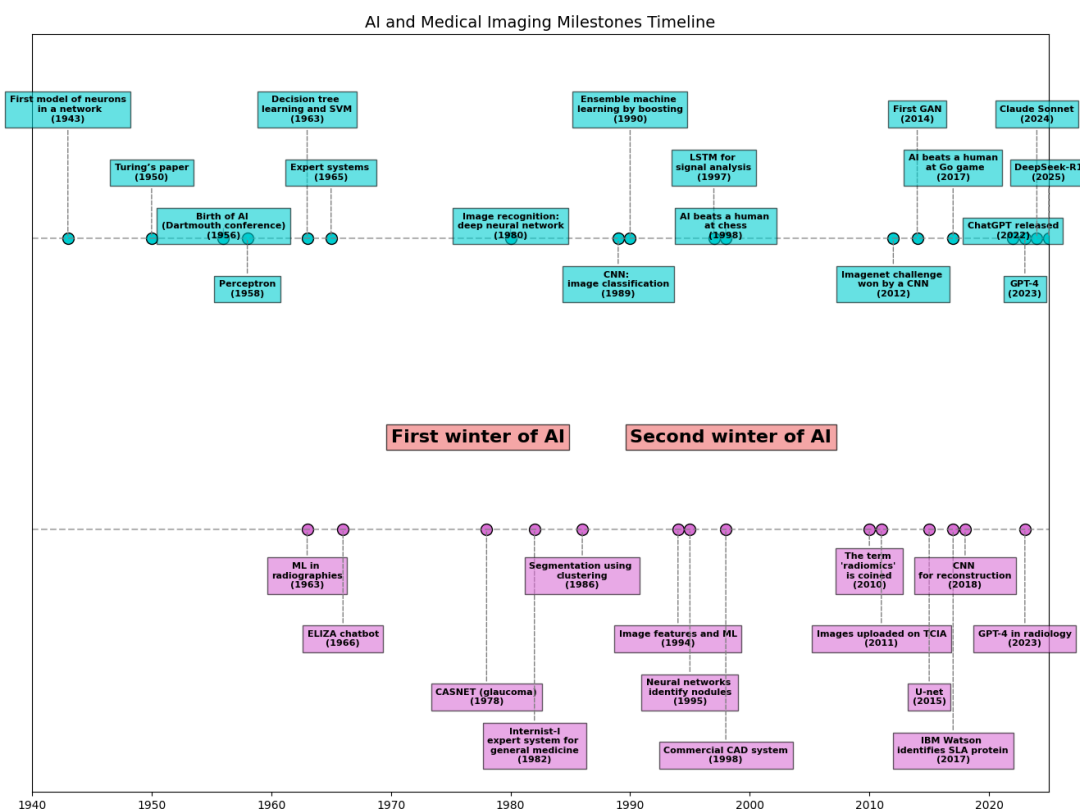


Figure 1-2 AI and machine learning in medical imaging brief timeline.

465 for being used in a semi-supervised strategy [Shah et al., 2018].

466 Many medical images datasets however, contain a high variability in class sizes
467 and variations in colors, which is specially noticeable in histopathological images
468 because of the usage of different staining and other factors which can affect the
469 color of the images (see section 2.2.3). This variability can lead to a significant loss
470 of efficiency of machine learning models when using a mixed supervision strategy,
471 as the model can be biased towards the most common classes or colors in the
472 dataset [Shah et al., 2018].

473 This is where other solutions arise to tackle the problem of the weak image
474 annotation while maintaining low costs. One of these solutions is crowdsourcing
475 strategy, which consists of having multiple annotators labeling the same image,
476 and then combining the labels to obtain a consensus label [Lu et al., 2023]. This
477 strategy can lead to a labeling cost reduction when different levels of expertise are
478 combined, since the crowd may be composed of both experts and laymen, being
479 the latter less expensive to hire [López-Pérez et al., 2023].

480 Recently, diagnosis, prognosis and treatment of cancer have heavily relied on
481 histopathology, where tissue samples are obtained through biopsies or surgical
482 resections and critical information that helps pathologists determine the presence
483 and severity of malignancies [López-Pérez et al., 2024]. The segmentation of
484 histopathological images enables precise identification of structures such as
485 nuclei, glands, and tumors, which are essential for assessing disease progression
486 and treatment response [Rashmi et al., 2021]. Accurate segmentation is
487 particularly crucial in digital pathology, where whole-slide images (WSI) are
488 analyzed using AI-powered CAD systems to support clinical decision-making
489 [López-Pérez et al., 2024].

490 A major challenge in histopathological image segmentation arises from the
491 variability in annotations provided by different pathologists. Unlike natural
492 images, where object boundaries are often well-defined, histological structures
493 may have ambiguous borders, leading to inconsistencies among annotators

[López-Pérez et al., 2023]. Because of this, crowdsourcing labeling is one of the most popular approaches, as illustrated in Figure 1-3, an example of how histopathological images are segmented by multiple experts, showing some variations in label assignment ¹. These discrepancies highlight the need for models that can handle annotation uncertainty effectively. Leveraging crowdsourcing strategies and machine learning techniques that infer annotator reliability can enhance segmentation performance while reducing costs.

1.2 Problem Statement

Throughout the development of medical technology and CAD, the task of ISS has become a crucial step in delivering precise diagnosis and treatment planning [Giri and Bhatia, 2024]. Particularly, in the area of histopathological studies, the usage of Whole Slide Images (WSI) is rather common since this method delivers high quality imaging and allows for the diagnosis of diseases like cancer [Lin et al., 2024].

ISS task consists of assigning a label to each pixel in an image according to the object it belongs to. Accurate segmentation is essential for the development of CAD systems, as it allows the identification of regions of interest (ROI) in the images, which can be used to detect and classify diseases and hence, treatment planning [Sarvamangala and Kulkarni, 2022]. However, modern computational solutions for ISS tasks involve the use of deep learning, which mostly rely large amounts of labeled data to train the models on supervised learning techniques. This means that the model is trained on a dataset with ground-truth labels, which are assumed to be correct and consistent across all samples. In practice, this assumption is often violated due to the high technical complexity of labeling these segments ².

¹obtained from a real world Triple Negative Breast Cancer (TNBC) dataset published in [López-Pérez et al., 2023]

²compared to a more trivial task like image classification on ordinary an well known classes like MNIST

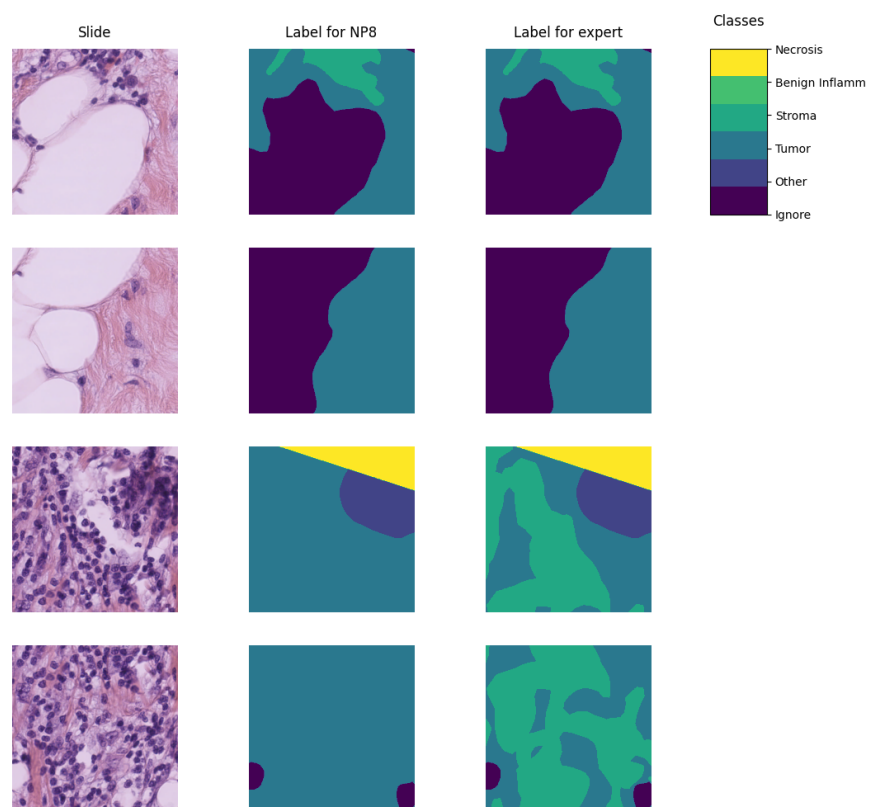


Figure 1-3 Example of a histopathological image segmented by multiple annotators, illustrating variations in label assignment.

The process of labeling medical images is often managed with the help of specialized software tools that allow the annotators to draw the regions, delivering an standard format for the labeled masks [Habis, 2024]. Despite the help of these tools, the labeling process in WSI can have high costs, as it requires long hours of work from specialized personnel. Because of cost constraints in many medical institutions, the labeling processes is often done by multiple labelers with varying levels of expertise, equalizing the cost of the labeling process. However, this strategy can lead to inconsistent labels, as the consensus between the labelers may not be exact due to the diversity in depth of knowledge and experience of the labelers [Xu et al., 2024]. These inconsistencies are mostly represented in the subsections 1.2.1 and 1.2.2.

1.2.1 Variability in Expertise Levels

One of the primary sources of inter-observer variability in medical image segmentation is the difference in expertise levels among annotators [López-Pérez et al., 2023]. Experienced radiologists and pathologists tend to produce highly precise annotations, whereas novice labelers may introduce systematic biases due to their limited familiarity with subtle image features. Studies have demonstrated that annotation accuracy tends to improve with experience, yet medical institutions often rely on a mix of annotators to manage costs and workload distribution [Lu et al., 2023].

The training background of annotators and institutional guidelines play a crucial role in shaping labeling practices. Different medical schools and hospitals may adopt distinct segmentation protocols, leading to inconsistencies when datasets are combined from multiple sources [López-Pérez et al., 2023]. For example, some institutions may emphasize conservative delineation of tumor boundaries, while others adopt a more inclusive approach. Such variations contribute to systematic biases in medical image datasets [Banerjee et al., 2025].

Medical images frequently contain structures with ambiguous boundaries, making segmentation inherently subjective. For instance, tumor margins in histopathological slides may not have well-defined edges, leading to variations in how different annotators delineate the regions of interest [Carmo et al., 2025]. These discrepancies arise not only from technical expertise but also from differences in perception and interpretation.

1.2.2 Technical Constraints and Image Quality

Technical constraints in medical imaging, such as resolution differences, noise levels, and contrast variations, can significantly impact segmentation accuracy. Lower-resolution images may obscure fine structures, leading to inconsistencies in boundary delineation [Zhou et al., 2024].

When combined with long sessions, bad images might also increase the cognitive load of the annotators, leading to fatigue and reduced precision in labeling [Kim et al., 2024]. This is particularly relevant in histopathological studies, where the staining process and tissue preparation can introduce color variations and artifacts that affect image quality, even if the same scanning equipment is used [Karthikeyan et al., 2023].

1.2.3 Research Question

Given the challenges posed by inconsistent labels in medical image segmentation, this work aims to address the following research question:

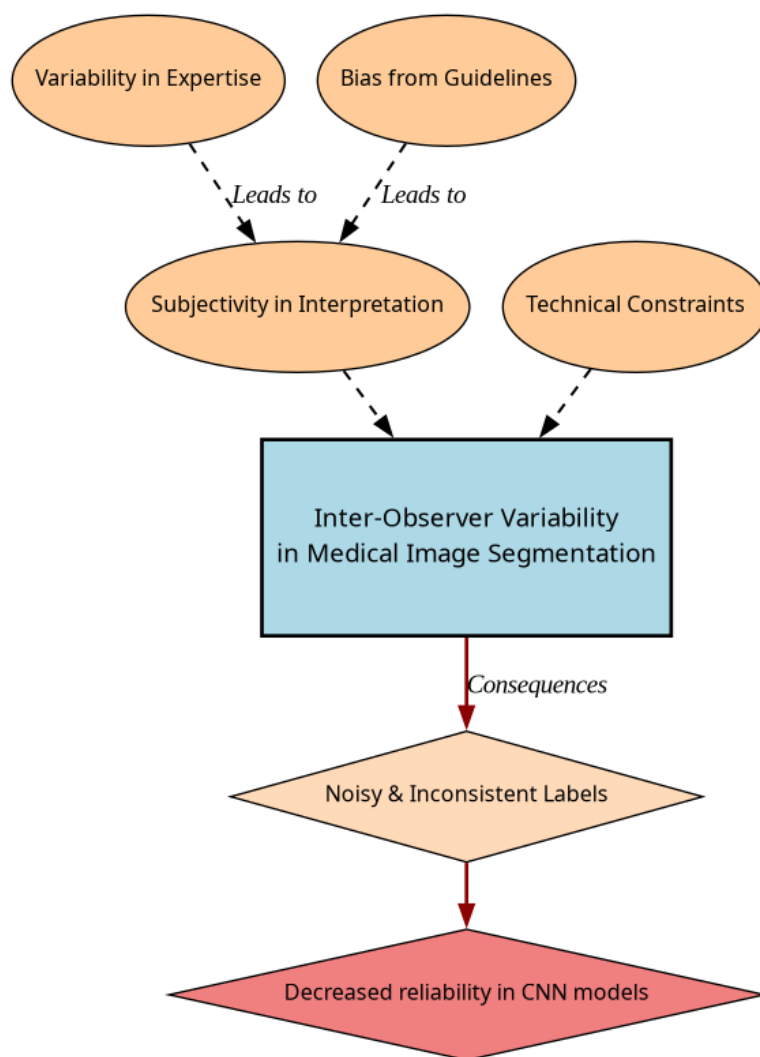


Figure 1-4 Summary diagram for problem Statement

Research Question

How can we develop a learning approach for ISS tasks in medical images that can adapt to inconsistent labels without requiring explicit supervision of labeler performance, while addressing challenges related to variability in expertise levels and technical constraints, and maintaining interpretability and generalization?

565

566 Certainly, in general **Machine Learning (ML)** classification tasks ³ where multiple
567 annotators are involved, **Majority Voting (MV)** is by far the simplest possible
568 approach to implement. This concept was born multiple times and divergently in
569 multiple fields, but it was described as relevant for **ML** and pattern recognition
570 labeling for classification in [Lam and Suen, 1997], in which the approach is
571 exposed as simple, yet powerful. The authors describe the **MV** as a method that
572 can be used to improve the accuracy of classification tasks by combining the labels
573 of multiple annotators. The method is based on the assumption that the majority
574 vote of the annotators is more likely to be correct than the vote of a single
575 annotator. The authors also describe the method as a straightforward way to
576 improve the accuracy of classification tasks without the need for complex
577 algorithms or additional data. The authors also prove this method to deliver very
578 similar results to more complicated approaches (Bayesian, logistic regression,
579 fuzzy integral, and neural network) in the particular task of **Optical Character**
580 **Recognition (OCR)**. Despite its simplicity, modern solutions for delivering accurate
581 medical image segmentation models still rely on Majority Voting at some stage,
582 like [Elnakib et al., 2020], which uses a majority voting strategy for delivering a
583 final output based on the labels of multiple models (VGG16-Segnet, Resnet-18 and
584 Alexnet) in **CT** images for Liver Tumor Segmentation, or [López-Pérez et al., 2023],
585 which uses **MV** for combining noisy annotations as an additional annotator to be
586 included in the deep learning solution. Majority voting as a technique for setting a
587 pseudo ground truth label is a powerful approach for its simplicity in many use

³In this work, image segmentation is considered as a particular case of classification in which target classes are assigned pixel-wise.

cases in which the target to be labeled is not tied to an expertise related task, otherwise, the assumption of equal expertise among the labelers can be a source of bias in the final label, which is not desirable in the case of highly technical annotations like medical images. In subsection 1.2.4, we will be reviewing literature which no longer assumes the naive approach of equal expertise among labelers and face the challenge of learning from inconsistent labels.

1.2.4 Facing Annotation Variability in Medical images

Learning from crowds approaches in general face the challenge of not having a ground truth label and hence, an intrinsic difficulty in measuring the real reliability of the labelers annotations. Some approaches assume beforehand a certain level of expertise for each labeler based on experience as an input, like in [TIAN and Zhu, 2015], which introduce the concept of max margin majority voting, using the reliability vector as weights for the weights for the binary and multiclass classifier. The crowdsourcing margin is the minimal difference between the aggregated score of the potential true label and the scores for other alternative labels. Accordingly, the annotators' reliability is estimated as generating the largest margin between the potential true labels and other alternatives. The problem introduced in this approach is assuming an stationary reliability per expert across the whole input space, which is imprecise since annotators performance may change between different tasks or even between different regions of the same image.

STAPLE Mechanism

The Simultaneous Truth and Performance Level Estimation (STAPLE) algorithm, introduced in [Warfield et al., 2004] is a probabilistic framework that estimates a hidden true segmentation from multiple segmentations provided by different raters. It also estimates the reliability of each rater by computing their sensitivity and specificity.

614 The **STAPLE** algorithm's goal is to maximize the log likelihood function:

$$(\mathbf{p}, \mathbf{q}) = \arg \max_{\mathbf{p}, \mathbf{q}} \ln f(\mathbf{D}, \mathbf{T} \mid \mathbf{p}, \mathbf{q}). \quad (1-1)$$

615 Where $\mathbf{D}$ is the set of segmentations provided by the raters, $\mathbf{T}$ is the hidden true
616 segmentation, p is the sensitivity and q is the specificity of the raters.

617 This is achieved by using the Expectation-Maximization algorithm to maximize the
618 log likelihood function in equation, which is done iteratively with step
619 computations:

$$\begin{aligned} (p_j^{(k)}, q_j^{(k)}) = \arg \max_{p_j, q_j} \sum_{i: D_{ij}=1} W_i^{(k-1)} \ln p_j \\ + \sum_{i: D_{ij}=1} (1 - W_i^{(k-1)}) \ln(1 - q_j) \\ + \sum_{i: D_{ij}=0} W_i^{(k-1)} \ln(1 - p_j) \\ + \sum_{i: D_{ij}=0} (1 - W_i^{(k-1)}) \ln q_j. \end{aligned} \quad (1-2)$$

620 The capacity of STAPLE to accurately estimate the true segmentation, even in the
621 presence of a majority of raters generating correlated errors, was demonstrated,
622 which makes it theoretically a strong choice for setting a ground-truth in binary or
623 multiclass medical **ISS** tasks.

624 The popularity and performance of **STAPLE** has led to its usage in modern
625 applications medical image, 3d spatial images due to its assumption of decision
626 space being based on voxel-wise decisions, like the authors in [Grefve et al., 2024]
627 which applied the algorithm on **Positron Emission Tomography (PET)** images.

628 Other authors still rely heavily on STAPLE for setting a ground truth consensus for
629 histopathological images, like [Qiu et al., 2022].

630 However, the STAPLE algorithm has some limitations. It assumes independent
631 rater errors, which may not hold in practice, leading to biased estimates. STAPLE
632 is also sensitive to low-quality annotations, potentially degrading final
633 segmentations if the weights are not initialized correctly. The algorithm tends to
634 over-smooth results, blurring fine details, and struggles with multi-class
635 segmentation. Computationally, it is expensive due to its iterative EM approach.
636 Additionally, STAPLE cannot correct systematic biases in annotations and depends
637 on initial estimates, impacting accuracy. Lastly, the estimated performance levels
638 lack interpretability, making it difficult to assess annotator reliability effectively.

639 Finally, this work contemplates STAPLE as useful for label aggregation, hence being
640 a good support for other methods, but not that useful for providing annotations of
641 structures on new and unlabeled images.

642 U-shaped CNNs

643 Since the introduction of U-Net [Ronneberger et al., 2015] in 2015 for biomedical
644 image segmentation, U-shaped CNNs have become a prevalent architecture in
645 medical image segmentation tasks. The U-Net’s success stems from its ability to
646 capture both global and local information through its contracting and expanding
647 paths, making it particularly effective for complex and heterogeneous structures,
648 even with limited annotated data. This architecture has been successfully applied
649 to various medical image segmentation tasks, including organ segmentation, tumor
650 segmentation, and brain structure segmentation.

651 The U-Net architecture consists of a symmetric encoder-decoder structure with
652 skip connections. The encoder path progressively reduces spatial dimensions
653 while increasing feature channels through a series of convolutional and
654 max-pooling layers, capturing high-level semantic information. The decoder path

655 uses transposed convolutions to gradually recover spatial resolution while
 656 reducing feature channels. Skip connections between corresponding encoder and
 657 decoder layers preserve fine-grained details by concatenating high-resolution
 658 features from the encoder with upsampled features in the decoder, enabling
 659 precise localization of structures.

660 In [López-Pérez et al., 2024] two networks are trained for delivering a final
 661 segmentation. One network is trained to estimate the annotators reliability and
 662 another one is trained to segment the image. The first network is a deep neural
 663 network that takes as input features of image and the labelers id encoded as
 664 one-hot and outputs a reliability map across the image feature space. This map is
 665 then used to weight the contribution of each annotator to the final segmentation.
 666 The second network is the U-Net used for segmentation.

667 In this approach, it is assumed that the images are labeled for at least one labeler
 668 and not all of them, which is closer to a real world scenario, in which it is common
 669 to have images with variability in the amount of annotations, per patch. Hence, the
 670 input data can be modeled as:

$$\mathcal{D} = (\mathbf{X}, \tilde{\mathbf{Y}}) = \{(\mathbf{x}_n, \tilde{\mathbf{y}}_n^r) : n = 1, \dots, N; r \in R_n\}, \quad (1-3)$$

671 Where every $\mathbf{x}_n$ is an input patch from a ROI in one WSI, $\tilde{\mathbf{y}}_n$ is the noisy annotation
 672 from the r labeler, N is the number of patches in the dataset and $R_n \subset \{1, \dots, R\}$
 673 is the set of labelers that annotated the image $\mathbf{x}_n$.

674 The authors then assume the annotator network to deliver a reliability map
 675 $\{\hat{\mathbf{A}}_\phi^{(r)}(\mathbf{x})\}_{r \in R_n}$ with different dimensions:

- 676 • CR global: a single reliability vector per labeler with dimensions C which
 677 represent global reliability of the labeler across all input space.

- CR image: a single reliability vector per image per labeler with dimensions C which represent local reliability of the labeler across the image.
- CR pixel: a reliability matrix per image per labeler, with dimensions C which represent local reliability of the labeler across all the pixels in the image.

These differences in dimensions are determined by the feature extraction space from segmentation network which feed the input of the annotator network, which the authors vary for experimentation purposes.

Being $\mathbf{p}_\theta(\mathbf{x}_n)$ the estimation of the latent (ground truth) segmentation delivered by the segmentation UNet network, thus, the estimated segmentation probability mask for each annotator is given by the product:

$$\mathbf{p}_{\theta,\phi}^{(r)}(\mathbf{x}_n) := \mathbf{A}_\phi^{(r)}(\mathbf{x}) \odot \mathbf{p}_\theta(\mathbf{x}_n), \quad (1-4)$$

where $\odot$ is the element-wise product and ϕ and θ are the parameters of the annotator network and the segmentation UNet network, respectively, being the latter initialized with a ResNet34 backbone pre-trained on ImageNet.

The authors propose a loss function involving cross-entropy and a trace based regularization on the reliability map, originally proposed in [Zhang et al., 2020] which combined, looks like:

$$\mathcal{L}(\theta, \phi) := \sum_{n=1}^N \sum_{r=1}^R \mathbb{I}(\tilde{\mathbf{y}}_n^{(r)} \in R_n) \cdot \left[\text{CE} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \cdot \mathbf{p}_\theta(\mathbf{x}_n), \tilde{\mathbf{y}}_n^{(r)} \right) + \lambda \cdot \text{tr} \left(\mathbf{A}_\phi^{(r)}(\mathbf{x}_n) \right) \right] \quad (1-5)$$

Being $\mathbb{I}$ the indicator function, CE the cross-entropy loss, and λ the regularization parameter.

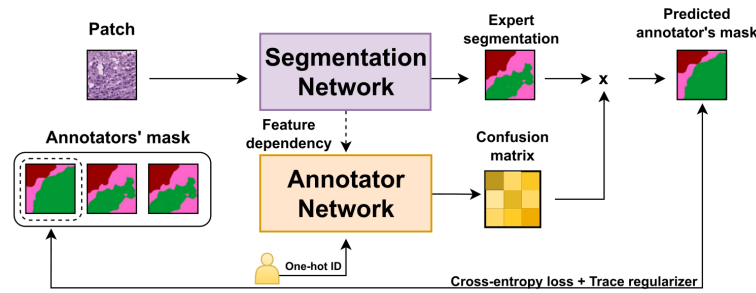


Figure 1-5 Proposed framework for the approach in [López-Pérez et al., 2024].

When evaluated on a Triple Negative Breast Cancer dataset, this approach achieves a Dice coefficient of 0.7827, outperforming STAPLE (0.7039) and matching expert-supervised performance (0.7723). The CR image reliability modeling proved most effective, as CR pixel, while potentially offering finer-grained reliability estimation, requires significantly more training data.

Despite the decent performance of the approach, solving the problem of multiple labelers with two networks can be overwhelming for the optimization process, requiring large amounts of annotated data to properly codify the annotators spatial reliabilities, which could be managed by a single model with an appropriate loss function.

Bayesian Models

Bayesian approaches are a good choice for handling label noise and uncertainty in the labelers. In [Julián and Álvarez Meza Andrés Marino, 2023] the authors propose a novel approach from Gaussian Processes to model the relationship between the annotators' reliability and the input data, while also preserving the interdependencies among the annotators. This is achieved by introducing **Correlated Chained Gaussian Processes for Multiple Annotators (CCGPMA)**, a framework based on the well known **Chained Gaussian Processes (CGP)**. CGP on itself cannot consider inter-annotator dependencies, thus, the authors introduce the **Correlated Chained Gaussian Processes (CCGP)** to model correlations between

the GP latent functions, which are supposed to be generated from a
Semi-Parametric Latent Factor Model (SLFM):

$$f_j(\mathbf{x}_n) = \sum_{q=1}^Q w_{j,q} \mu_q(\mathbf{x}_n), \quad (1-6)$$

where $f_j : \mathcal{X} \rightarrow \mathbb{R}$ is a Latent Function (LF), $\mu_q(\cdot) \sim \mathcal{GP}(0, k_q(\cdot, \cdot))$ with $k_q : \mathcal{X} \times \mathcal{X} \rightarrow \mathbb{R}$ being a kernel function, and $w_{j,q} \in \mathbb{R}$ is a combination coefficient ($Q \in \mathbb{N}$). This leads to a joint distribution of the form:

$$p(\mathbf{y}, \hat{\mathbf{f}}, u | \mathbf{X}) = p(\mathbf{y} | \boldsymbol{\theta}) \prod_{j=1}^J p(\mathbf{f}_j | \mathbf{u}) p(\mathbf{u}), \quad (1-7)$$

where $\mathbf{y}$ is the vector of noisy labels, $\hat{\mathbf{f}}$ is the vector of latent functions, u represents the inducing points, and $\mathbf{X}$ is the input data.

Combined with inducing-variables based methods for sparse GP approximations, and maximizing an Evidence Lower Bound (ELBO) for the estimation of the variational parameters, the authors reach a model whose variational expectations are not analytically tractable, and hence, the authors derive a Gaussian-Hermite quadrature approach.

Finally, the authors extend this approach for being applied to classification and regression, reaching the only known approach to involve chained gaussian processes in multiple annotators classification and regression tasks while

731 preserving the interdependencies among the annotators, and also outperforming
 732 GPC-MV⁴, MA-LFC-C⁵, MA-DGRL⁶, MA-GPC⁷, MA-GPCV⁸, MA-DL⁹, KAAR¹⁰.

733 CCGPMA on itself proposes a good approach for handling label noise and
 734 uncertainty in the labelers for regression and classification tasks, while also
 735 preserving the interdependencies among the annotators, however, it does not face
 736 the image segmentation problem, which is the main focus of this works, however,
 737 it does not face the image segmentation problem, which is the main focus of this
 738 work. Besides, handling so many latent functions during the optimization process
 739 is computationally expensive, making it on itself infeasible for large and high
 740 resolution datasets.

741 1.2.5 Facing Noisy Annotations and Low-Quality Data

742 The problem of low-quality data and noisy annotations has been tackled with
 743 various strategies. One such approach is the use of deep learning models that
 744 incorporate loss functions designed to mitigate the effects of unreliable labels.
 745 Traditional methods such as Majority Voting (MV) or Expectation-Maximization
 746 (EM) have been widely used for aggregating multiple annotators' inputs. However,
 747 they assume a homogeneous reliability of annotators, which may not hold in
 748 real-world scenarios.

⁴A GPC using the MV of the labels as the ground truth.

⁵A LRC with constant parameters across the input space.

⁶A multi-labeler approach that considers as latent variables the annotator performance.

⁷A multi-labeler GPC, which is an extension of MA-LFC.

⁸An extension of MA-GPC that includes variational inference and priors over the labelers' parameters.

⁹A Crowd Layer for DL, where the annotators' parameters are constant across the input space.

¹⁰A kernel-based approach that employs a convex combination of classifiers and codes labelers dependencies.

Loss Functions in Deep Learning Models

Loss functions are fundamental components in deep learning models that quantify how well a model's predictions match the ground truth. They serve as the objective function that guides the learning process by measuring the discrepancy between predicted and actual values. In classification tasks, the most common loss functions are Cross-Entropy (CE) and Mean Absolute Error (MAE) [Zhang and Sabuncu, 2018]. CE is particularly effective for classification as it heavily penalizes confident but wrong predictions, though it can be sensitive to noisy labels. MAE, on the other hand, is more robust to outliers and assigns equal weights to all mistakes, but typically requires more training iterations. For image segmentation tasks, specialized loss functions have been developed to handle the unique challenges of pixel-wise classification. The Dice loss, which measures the overlap between predicted and ground truth regions, is widely used in medical image segmentation [Zhao et al., 2020]. More recently, the Generalized Cross Entropy (GCE) loss has emerged as a robust alternative that combines the benefits of both CE and MAE, allowing for better handling of noisy labels through a tunable parameter that controls sensitivity to outliers. However, to the best of our knowledge, none of the current loss functions approaches have focused on the multiple labelers scenario, neither on the per labeler spatial reliability estimation.

Generalized Cross-Entropy for Multiple Annotators Classification

A more recent approach was proposed by [Triana-Martinez et al., 2023], introducing a Generalized Cross-Entropy-based Chained Deep Learning (GCECDL) framework. This method addresses the limitations of traditional label aggregation techniques by modeling each annotator's reliability as a function of the input data. The approach effectively mitigates the impact of noisy labels by using a noise-robust loss function, balancing Mean Absolute Error (MAE) and Categorical Cross-Entropy (CE). Unlike prior approaches, GCECDL accounts for the dependencies among annotators while encoding their non-stationary behavior

777 across different data samples. Their experiments on multiple datasets
 778 demonstrated superior predictive performance compared to state-of-the-art
 779 methods, particularly in cases where annotations were highly inconsistent.

780 The strategy of the authors effectively unlocks the potential of **ML** models to handle
 781 low-quality data and noisy annotations, but it is bounded to classifications tasks
 782 only, not being by itself applicable to segmentation tasks. The TGCE equation for
 783 handling multiple annotators is defined as:

$$\text{TGCE}(\mathbf{y}, f(\mathbf{x}); \tilde{\lambda}_x, \tilde{C}) = \tilde{\lambda}_x \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q} + (1 - \tilde{\lambda}_x) \frac{1 - (\tilde{C})^q}{q}, \quad (1-8)$$

784 where $\tilde{\lambda}_x$ represents the annotator reliability, $\tilde{C}$ is a constant, q is a parameter that
 785 controls the balance between MAE and CE behavior, $\mathbf{y}$ is the annotation vector, and
 786 $f(\mathbf{x})$ is the model prediction. This approach is more deeply discussed in chapter 4.

787 1.3 Aims

788 1.3.1 General Aim

789 The main purpose of this work is to develop a novel approach for **ISS** tasks in
 790 medical images, which can adaptively infer the best possible segmentation without
 791 needing separate inputs about the labeler offering a much more interpretable
 792 reliability space per labeler than other solutions in the literature and correctly
 793 facing the performance inconsistency across the annotators space and the
 794 variability of images quality.

Table 1-1 Summary of state-of-the-art approaches for handling multiple annotators in medical image segmentation

Approach	Key Characteristics	Advantages	Limitations
Majority Voting (MV) [Lam and Suen, 1997]	Simple aggregation of multiple annotators' labels	- Simple to implement - No complex algorithms required - Proven effective in many cases	- Assumes equal expertise - Sensitive to correlated errors - May introduce bias in expert tasks
STAPLE [Warfield et al., 2004]	Probabilistic framework estimating true segmentation and rater reliability	- Handles binary and multiclass segmentation - Estimates rater sensitivity/specificity - Robust to correlated errors	- Computationally expensive - Sensitive to low-quality annotations - Tends to over-smooth results - Limited interpretability
U-Net based approaches [Ronneberger et al., 2015]	Deep Learning architecture with encoder-decoder structure	- Captures both global and local information - Effective for complex structures - Works well with limited data	- Requires large amounts of training data - Complex optimization process - May need specialized loss functions
Bayesian Models (CCGPMA) [Gil-Gonzalez et al., 2023]	Gaussian Processes modeling annotator reliability	- Preserves annotator interdependencies - Handles label noise effectively - Works for classification and regression	- Computationally expensive - Not directly applicable to segmentation - Complex optimization process
Generalized Cross-Entropy (GCE) [Zhang and Sabuncu, 2018]	Loss function balancing MAE and CE	- Robust to noisy labels - Accounts for non-stationary behavior - Handles annotator dependencies	- Limited to classification tasks - Requires parameter tuning - May need adaptation for segmentation

1.3.2 Specific Aims

- To develop a novel loss function for ISS tasks in medical images, capable of inferring the best possible segmentation without needing separate inputs about the labeler performance.
- Introducing a tensor map which codifies the reliability of each labeler, allowing the model to implicitly weigh the labelers based on their performance across the mask and classes space.
- To develop and test a deep learning model for ISS tasks in medical images, which can learn from inconsistent labels and improve the segmentation performance compared to other solutions in state of the art.

1.4 Outline and Contributions

Building upon the considerations discussed in Section 1.2, this thesis introduces a novel approach to ISS in medical imaging. The core of this work is a model designed to adapt its learning process to varying labeler performance. This is achieved through a unique loss function that infers optimal segmentation and assesses labeler reliability without requiring explicit performance inputs. This loss function implicitly weighs labelers based on their performance, utilizing an intermediate reliability map. This map enables the model to prioritize information from reliable labelers while disregarding noisy labels. Consequently, this methodology diverges from conventional CNN-based segmentation models by eliminating the need for direct supervision of labeler performance, thereby enhancing its generalizability and adaptability across diverse datasets and labeler cohorts.

The primary contributions of this research to the field of ISS in medical images are as follows:

- The development of a novel loss function for ISS tasks. This function can infer the best possible segmentation and codify labeler reliability without needing separate inputs regarding labeler performance or the training of multiple networks (see Chapter 4).
- The creation of a Chained Deep Learning model for ISS in medical images. This model effectively learns from inconsistent labels and demonstrates improved segmentation performance, outperforming state-of-the-art approaches on datasets with emulated noisy labels and a real world Triple Negative Breast Cancer histopathology dataset (see Chapter 5).
- A mechanism for emulating noisy annotations. This involves using network encoding layer weight disturbances to construct a crowdsourced version of the well-known Oxford-IIIT Pet Dataset (see Section 5.1.1).

- 832 • A Python package facilitating the use of the proposed loss function and
833 architecture within CNN models for ISS tasks in medical images.¹¹
- 834 • Mappings of the Triple Negative Breast Cancer and Oxford-IIIT Pet Datasets as
835 lazy loaders, integrated into the package for immediate use with the proposed
836 loss function.¹²
- 837 • A publicly accessible Github repository containing the source code developed
838 for this work.¹³

¹¹https://pypi.org/project/seg_tgce/

¹²<https://seg-tgce.readthedocs.io/en/latest/experiments.html>

¹³https://github.com/blotero/seg_tgce

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CHAPTER

TWO

843

CONCEPTUAL PRELIMINARIES

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2.1 Modern Concept of Digital Image

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A digital image is a numerical representation of a visual scene, captured through various imaging devices and stored in a computer. From a mathematical perspective, a digital image can be represented as a function $f(x, y)$ that maps spatial coordinates (x, y) to intensity values. In the discrete domain, this function is sampled at regular intervals, creating a matrix of values known as pixels (picture elements).

850

2.1.1 Types of Digital Images

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Grayscale Images

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Grayscale images are the simplest form of digital images, where each pixel represents a single intensity value. Mathematically, a grayscale image can be represented as a 2D matrix I of size $M \times N$, where each element $I(i, j)$ represents the intensity at position (i, j) . The intensity values typically range from 0 (black) to 255 (white) in 8-bit images, or from 0 to 65535 in 16-bit images.

857 **Color images**

858 Color images extend the grayscale concept by representing each pixel with multiple
859 channels, typically Red, Green, and Blue (RGB). A color image can be represented
860 as a 3D matrix I of size $M \times N \times 3$, where $I(i, j, k)$ represents the intensity of the
861 k -th color channel at position (i, j) . Other color spaces like HSV (Hue, Saturation,
862 Value) or CMYK (Cyan, Magenta, Yellow, Key) are also commonly used in different
863 applications.

864 **Multispectral images**

865 Multispectral images capture information across multiple wavelength bands
866 beyond the visible spectrum. These images can be represented as a 3D matrix I of
867 size $M \times N \times B$, where B is the number of spectral bands. Each band $I(i, j, b)$
868 represents the intensity at position (i, j) for the b -th spectral band. This
869 representation is particularly useful in medical imaging, remote sensing, and
870 scientific applications.

871 **3D Images and Volumetric Data**

872 Three-dimensional images extend the concept of pixels to voxels (volume elements).
873 A 3D image can be represented as a 3D matrix V of size $M \times N \times D$, where D
874 represents the depth dimension. Each voxel $V(i, j, k)$ represents the intensity at
875 position (i, j, k) in the 3D space. This representation is fundamental in medical
876 imaging (CT, MRI), scientific visualization, and computer graphics.

877 **2.1.2 Mathematical Representations**

878 The mathematical foundation of digital images relies on several key concepts:

- 879 • **Sampling:** The process of converting a continuous image into a discrete
880 representation. According to the Nyquist-Shannon sampling theorem, the
881 sampling frequency must be at least twice the highest frequency present in
882 the image to avoid aliasing.
- 883 • **Quantization:** The process of converting continuous intensity values into
884 discrete levels. The number of quantization levels determines the image's
885 bit depth and affects its quality and storage requirements.
- 886 • **Resolution:** The number of pixels per unit length in an image, typically
887 measured in pixels per inch (PPI) or dots per inch (DPI).
- 888 • **Dynamic range:** The ratio between the maximum and minimum measurable
889 light intensities in an image, often expressed in decibels (dB).

890 The mathematical representation of a digital image can be expressed as:

$$I(x, y) = \sum_{i=0}^{M-1} \sum_{j=0}^{N-1} f(i, j) \cdot \delta(x - i, y - j) \quad (2-1)$$

891 where $I(x, y)$ is the digital image, $f(i, j)$ represents the intensity values, and $\delta(x -$
892 $i, y - j)$ is the Kronecker delta function.

893 For color images, the representation extends to:

$$I(x, y) = \begin{bmatrix} I_R(x, y) \\ I_G(x, y) \\ I_B(x, y) \end{bmatrix} \quad (2-2)$$

894 where I_R , I_G , and I_B represent the red, green, and blue channels respectively.

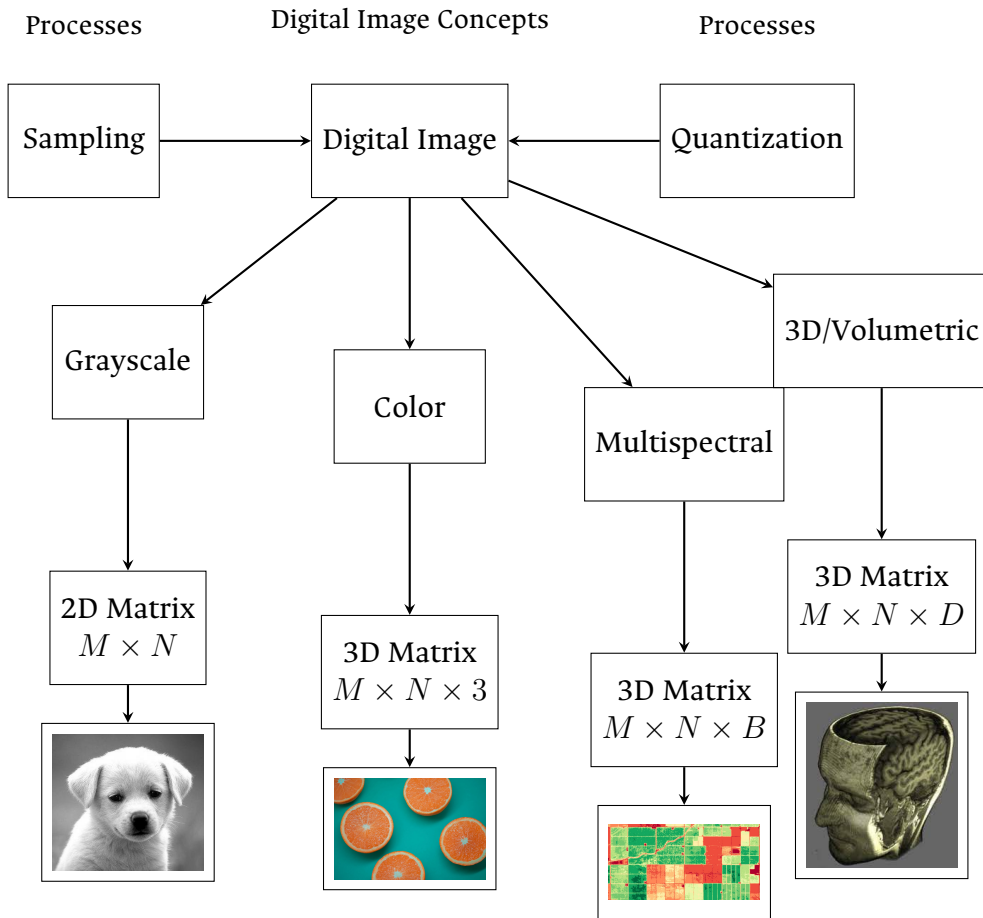


Figure 2-1 Overview of digital image concepts and their mathematical representations. The figure shows the main types of digital images (grayscale, color, multispectral, and volumetric), their mathematical representations, and the fundamental processes of sampling and quantization. Example images are included to illustrate each type.

2.2 Digital Histopathological Images

Digital histopathology represents a significant advancement in medical imaging, where traditional glass slides containing tissue samples are digitized using specialized scanning devices. This transformation has revolutionized the field of pathology by enabling remote diagnosis, computer-aided analysis, and digital archiving of tissue samples [Amgad et al., 2019]. This process has evolved significantly over the past few decades, as shown in Figure 2-2.

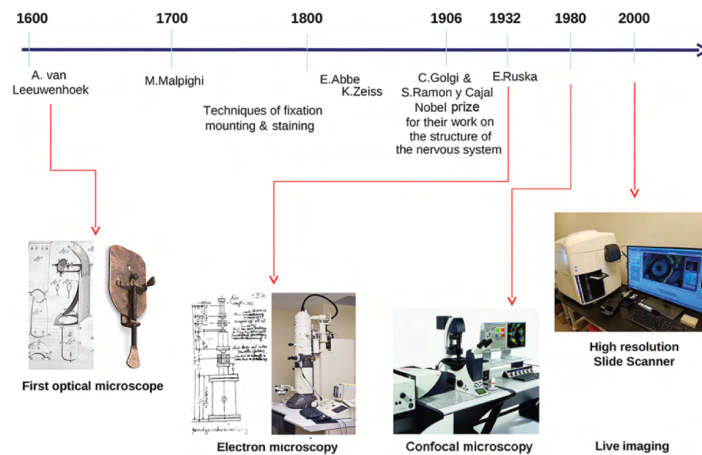


Figure 2-2 Histology evolution timeline. (Image from [Mazzarini et al., 2021]).

2.2.1 Whole Slide Imaging (WSI)

Whole Slide Imaging (WSI) is the process of digitizing entire glass slides at high resolution, creating a digital representation that can be viewed, analyzed, and shared electronically. Modern WSI scanners use sophisticated optical systems that capture multiple fields of view at high magnification, which are then stitched together to create a seamless digital image [Hu et al., 2025]. These systems incorporate high-resolution objectives with magnifications ranging from 20x to 40x, precise motorized stages for accurate slide positioning, automated focus

910 systems to maintain image quality, and high-quality cameras equipped with large
911 sensor arrays. The resulting digital slides can reach sizes of several gigabytes,
912 containing billions of pixels that capture the microscopic details of tissue samples
913 [Hu et al., 2025]. Figure 2-3 shows a whole slide imaging system by Omnyx for
914 slide digitization and a comprehensive digital pathology interface from Omnyx
915 designed to streamline pathologists' diagnostic workflow [Farahani et al., 2015].

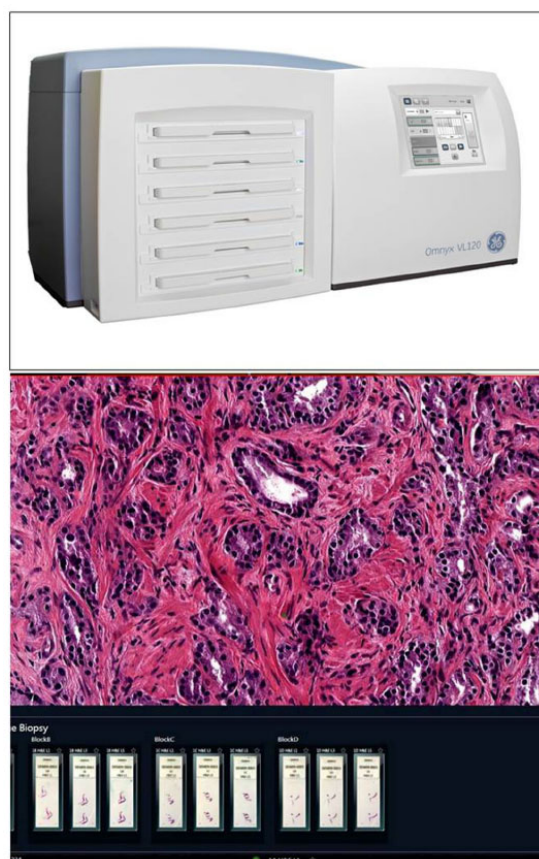


Figure 2-3 (Above) Whole slide imaging system by Omnyx for slide digitization. (Below) Comprehensive digital pathology interface from Omnyx designed to streamline pathologists' diagnostic workflow. (From [Farahani et al., 2015]).

916 2.2.2 Regions of Interest (ROI)

917 In digital histopathology, **Region of Interests (ROIs)** are specific areas within a
918 whole slide image that contain diagnostically relevant information. These regions
919 can be manually annotated by pathologists, automatically detected using
920 computer vision algorithms, or defined based on specific tissue characteristics or
921 abnormalities. The importance of **ROIs** lies in their ability to focus computational
922 analysis on relevant areas, reduce computational complexity in automated
923 systems, facilitate targeted diagnosis and research, and enable efficient storage and
924 transmission of critical information.

925 2.2.3 Staining Techniques

926 Histopathological analysis relies heavily on various staining techniques to enhance
927 the visibility of different tissue components and cellular structures. The choice of
928 staining method depends on the specific diagnostic requirements and the type of
929 tissue being examined.

930 Hematoxylin and Eosin (H&E)

931 Hematoxylin and Eosin (H&E) staining is the most widely used technique in
932 histopathology, particularly in breast cancer diagnosis [Pan et al., 2021]. This
933 staining method provides essential visualization through two components:
934 hematoxylin, which stains cell nuclei blue/purple to highlight nuclear morphology,
935 and eosin, which stains cytoplasm and extracellular matrix pink/red to reveal
936 tissue architecture.

937 The popularity of H&E staining in breast cancer histopathology stems from its
938 ability to clearly visualize tumor architecture and growth patterns, distinguish
939 between different types of breast cancer, identify important diagnostic features

like nuclear pleomorphism, and assess tumor grade and stage. Beyond breast cancer, H&E staining finds extensive application across various medical specialties including general pathology, dermatology, gastroenterology, neurology, and oncology.

Special Stains

In addition to H&E, various special stains are used for specific diagnostic purposes. Immunohistochemistry (IHC) uses antibodies to detect specific proteins, playing a crucial role in subtyping breast cancers. Key IHC stains include Estrogen Receptor (ER) staining for detecting estrogen receptors, Progesterone Receptor (PGR) staining for assessing progesterone receptor status, Human Epidermal Growth Factor Receptor 2 (HER2) staining for evaluating HER2 protein expression, and Ki67 staining for measuring cellular proliferation rates. These markers are particularly crucial in breast cancer diagnosis and treatment planning, as they help determine the molecular subtype of the cancer and guide personalized therapeutic approaches. Other specialized stains include Periodic Acid-Schiff (PAS) for highlighting carbohydrates and basement membranes, Masson's Trichrome for distinguishing between collagen and muscle fibers, and silver stains for detecting microorganisms and nerve fibers. These specialized staining techniques complement H&E by providing additional diagnostic information that is crucial for accurate diagnosis and treatment planning [Weitz et al., 2023]. Examples of these staining techniques are shown in Figure 2-7.

2.3 Deep Learning Fundamentals

Deep Learning has emerged as a powerful subset of machine learning, revolutionizing the field of artificial intelligence. Its roots can be traced back to the early development of artificial neural networks in the 1940s and 1950s, with significant milestones including the perceptron in 1958 and the backpropagation

algorithm in the 1980s. However, it wasn't until the early 21st century, with the advent of more powerful computational resources and the availability of large datasets, that deep learning truly began to flourish.

2.3.1 Learning Paradigms

Deep Learning systems can be categorized into three main learning paradigms. The most common approach is supervised learning, where models learn from labeled data by mapping inputs to known outputs. This paradigm requires a large amount of labeled training data, which can be expensive and time-consuming to acquire. Semi-supervised learning offers a hybrid approach that leverages both labeled and unlabeled data, proving particularly useful when labeled data is scarce but unlabeled data is abundant. Finally, unsupervised learning enables models to discover patterns and structures from unlabeled data without explicit guidance, making it valuable for tasks like clustering and dimensionality reduction.

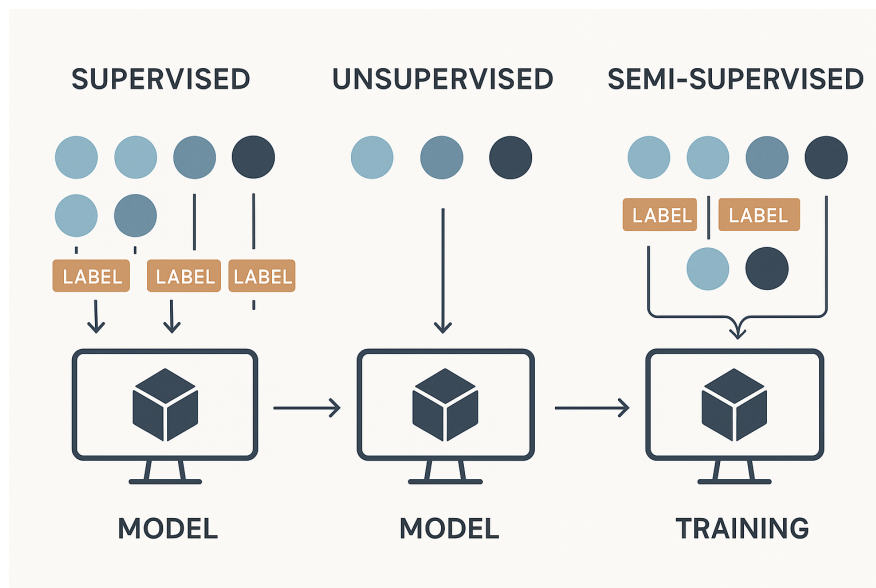


Figure 2-4 Common learning paradigms.

979 2.3.2 Architecture and Training

980 Deep Learning architectures are characterized by their layered structure, where
981 each layer progressively extracts and transforms features from the input data. The
982 early layers typically focus on low-level feature extraction, such as edges, textures,
983 and basic patterns in the case of image processing. As information flows through
984 the network, middle layers combine these basic features into more complex
985 representations. The final layers perform high-level reasoning and make the
986 ultimate predictions or classifications.

987 The training process relies heavily on the gradient descent algorithm, which
988 iteratively adjusts the model's parameters to minimize a loss function. This loss
989 function serves as a crucial component of the learning process, quantifying how
990 well the model's predictions match the actual targets. By providing a measure of
991 the model's performance, the loss function guides the optimization process,
992 enabling the network to learn meaningful patterns from the training data.

993 2.3.3 Challenges and Solutions

994 Despite their power, deep learning systems face several significant challenges. One
995 of the most prominent issues is overfitting, where models may memorize training
996 data instead of learning generalizable patterns. This challenge is typically
997 addressed through various regularization techniques such as dropout, L1/L2
998 regularization, and early stopping. Another critical challenge is the substantial
999 data requirements; deep learning models often need massive amounts of training
1000 data to achieve good performance, which can be a limiting factor in many
1001 applications. Additionally, the complex, layered nature of deep learning models
1002 makes them difficult to interpret, often referred to as "black boxes." This lack of
1003 transparency can be particularly problematic in critical applications where
1004 understanding the decision-making process is essential.

2.3.4 Deep Learning Frameworks

The development of powerful open-source frameworks has significantly accelerated deep learning research and applications. TensorFlow, developed by Google, provides a comprehensive ecosystem for building and deploying machine learning models [Abadi et al., 2016]. PyTorch, created by Facebook’s AI Research lab, offers dynamic computation graphs and has become particularly popular in research settings. Caffe, known for its speed and modularity, is widely used in computer vision applications.

These frameworks have democratized deep learning by providing efficient implementations of common operations, automatic differentiation for gradient computation, and GPU acceleration for faster training. They also offer pre-trained models and transfer learning capabilities, along with active communities for support and knowledge sharing. The combination of these frameworks with modern hardware has enabled researchers and practitioners to develop increasingly sophisticated models, pushing the boundaries of what’s possible in artificial intelligence. As shown in Figure 2-5, which presents data from Google Trends over the last five years (as of April 2025), TensorFlow and PyTorch have emerged as the two most prominent frameworks in the deep learning landscape.

2.4 Datasets and Data Sources

Throughout the development of this work, multiple datasets were used for evaluation of ISS models. The common elements of all these datasets are that they contain RGB images and are crowdsourced with multiple labelers, where not necessarily all labeler label all images.

As it has been mentioned in Chapter 1, the main goal of this work is mainly focused on crowdsourced histopathology images semantic segmentation, however, these datasets present the following challenges:

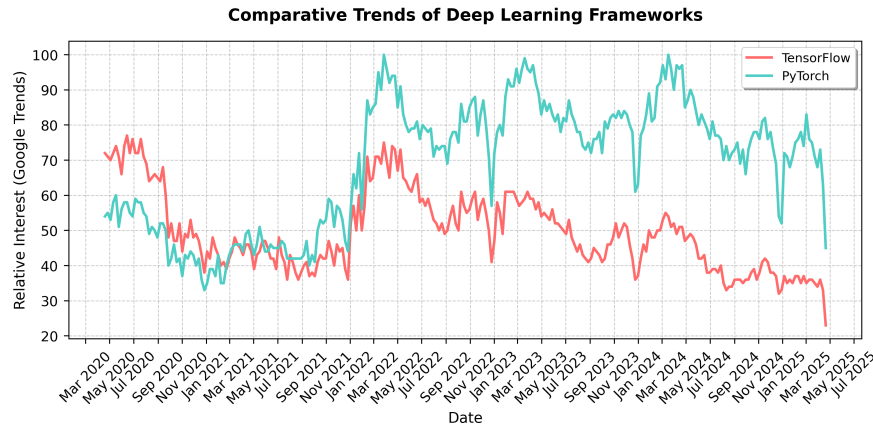


Figure 2-5 Comparative Trends of the top two most popular Deep Learning Frameworks, apparently, tendency was switched to PyTorch since 2022

- 1031 • Distribution of segmentation labels is not uniform across the image, since
- 1032 some tissues and structures are more common than others.
- 1033 • Visualization of performance of the models in debug time (like per epoch
- 1034 analysis) is not simple for non experts in the subject, which makes it hard to
- 1035 evaluate whether the model is overfitting or not at a glance.

1036 For these reasons, multiple datasets were created in the pursuit of an initial
 1037 evaluation of performance of the models against more traditional and familiar
 1038 images before the focus on histopathology images. Once a decent performance in
 1039 metrics like Dice coefficient was achieved, the focus was shifted to histopathology
 1040 images and further tunings on the models were performed if needed.

1041 In any case, both the emulated noisy annotations datasets and the histopathology
 1042 datasets somehow contained ground truth aggregation, either from the original
 1043 source (in the case of emulated noisy annotations), the expert annotation (if
 1044 available) or from the aggregation of multiple labelers ¹.

¹STAPLE in the case of histopathology datasets with no expert annotations available

2.4.1 Oxford-IIIT Pet Dataset

Using the techniques described later in Section 5.1.1, the Oxford-IIIT Pet Dataset [Parkhi et al., 2012] was used to create a dataset with emulated noisy annotations. This dataset is officially available in the ‘tensorflow_datasets’ package, making it convenient to use in experiments. It is composed of 7349 images of pets, being 2371 of these images cats, and the rest 4978, dogs. Breed labels are included as well as segmentation masks for the classes: ‘background’, ‘border’, and ‘pet’. Figure 2-6 shows an example of the annotations in the original dataset.

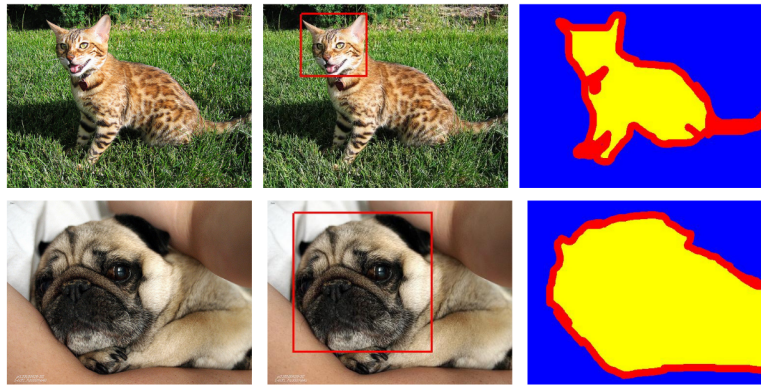


Figure 2-6 Annotations in the Oxford-IIIT Pet data. From left to right: pet image, head bounding box, and trimap segmentation (blue: background region; red: ambiguous region; yellow: foreground region).

2.4.2 Real histopathology datasets

Multi-Stain Breast Cancer Histological Dataset

The Multi-Stain Breast Cancer Histological Dataset [Weitz et al., 2023] represents one of the largest publicly available collections of whole slide images (WSIs) from surgical resection specimens of primary breast cancer patients. This dataset is particularly valuable for our work because it contains matched pairs of H&E and

IHC-stained tissue sections from the same tumor, with a total of 4,212 WSIs from 1,153 patients. The IHC stains include important biomarkers such as ER, PGR, HER2, and KI67, which are routinely used in breast cancer diagnosis and treatment planning (more on staining techniques in Section 2.2.3).

The dataset’s relevance to our work stems from several key aspects. The matched H&E and IHC stains allow for studying the consistency of segmentation across different staining modalities, which is crucial for understanding how different visualization methods affect annotation quality. With 1,153 patients, the dataset provides a robust foundation for training and evaluating segmentation models in a real-world clinical setting. The inclusion of routine diagnostic cases makes the dataset representative of actual clinical practice, where variations in staining quality and tissue preparation are common. Furthermore, the multiple biomarker stains (ER, PGR, HER2, KI67) enable the study of how different tissue characteristics affect segmentation performance and annotator agreement.

This dataset serves as an ideal testbed for our crowdsourced segmentation approach. It allows us to evaluate how different staining modalities affect annotator performance and agreement, while also providing insights into the relationship between tissue characteristics and segmentation difficulty. The dataset’s comprehensive nature enables validation of our models’ performance across different biomarker expressions and assessment of the generalizability of our approach to real-world clinical data.

Structured Crowdsourcing Dataset for Histology Images

The dataset presented by [Amgad et al., 2019] is particularly relevant to our work as it represents one of the first systematic studies of crowdsourced annotations in histopathology. The authors recruited 25 participants with varying levels of expertise (from senior pathologists to medical students) to delineate tissue regions in 151 breast cancer slides using the Digital Slide Archive platform, resulting in over 20,000 annotated tissue regions.

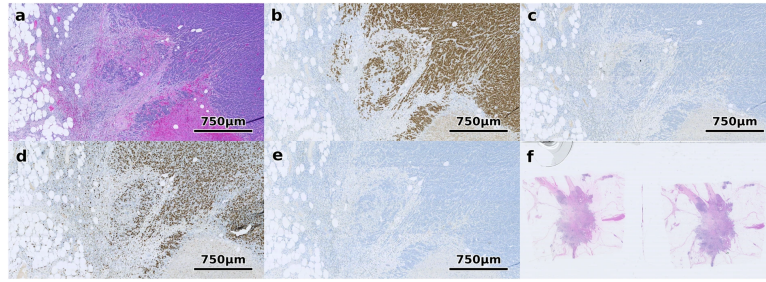


Figure 2-7 Different staining techniques obtained from multi-stain breast cancer dataset [Weitz et al., 2023]. (a) shows H&E, (b) ER, (c) HER2, (d) Ki67 and (e) PGR. (f) shows an example of a WSI that was excluded since it contains multiple tissue sections.

Key aspects of this dataset make it valuable for our work. The systematic evaluation of inter-participant discordance revealed varying levels of agreement across different tissue classes, with low discordance for tumor and stroma, and higher discordance for more subjectively defined or rare tissue classes. The inclusion of feedback from senior participants helped in curating high-quality annotations, demonstrating that fully convolutional networks trained on these crowdsourced annotations can achieve high accuracy (mean AUC=0.945). The dataset also provides evidence that the scale of annotation data significantly improves image classification accuracy.

This dataset is particularly valuable for our work because it provides crucial insights into how annotator expertise affects segmentation quality. It demonstrates the feasibility of using crowdsourced annotations for training accurate segmentation models, showing that even with varying levels of expertise, aggregated annotations can produce reliable ground truth. The dataset includes a systematic analysis of inter-annotator agreement, which is crucial for understanding the challenges in crowdsourced histopathology segmentation and informing the development of more robust segmentation approaches.

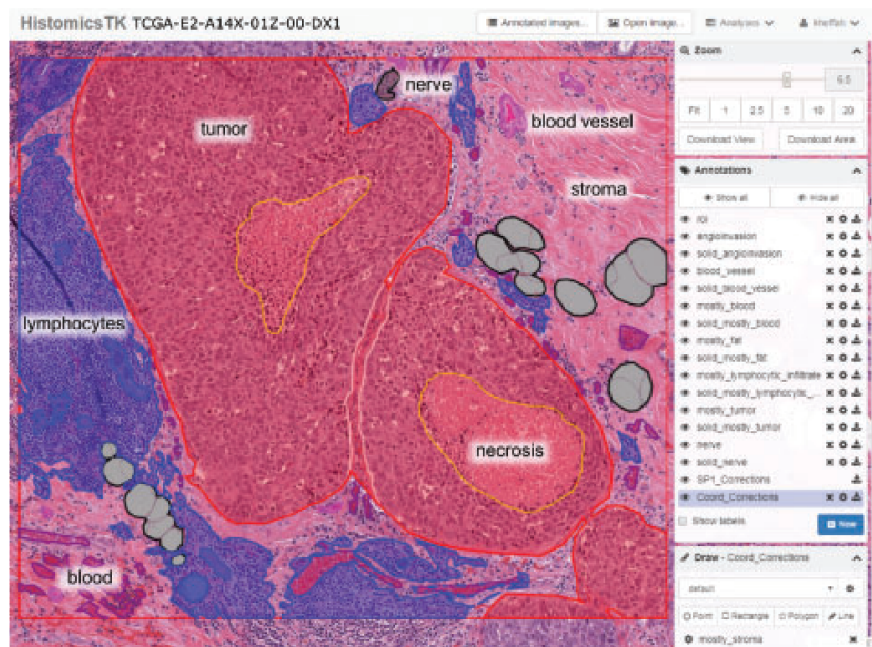


Figure 2-8 Screenshot of the DSA and HistomicsTK web interface while creating the crowdsourced annotations for the dataset presented by [Amgad et al., 2019].

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CHAPTER

1106

THREE

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1108

CHAINED GAUSSIAN PROCESSES

1109

3.1 Background and Related Methods

1110

In recent years, several approaches have been proposed to handle supervised learning problems in the context of multiple annotators. This section reviews key methods that form the foundation for understanding chained Gaussian processes and their application to multiple annotator scenarios in regression and classification tasks.

1114

1115

3.1.1 Kernel Alignment-Based Annotator Relevance Analysis (KAAR)

1116

1117

The Kernel Alignment-Based Annotator Relevance Analysis (KAAR) method, introduced in [Julián and Álvarez Meza Andrés Marino, 2023], addresses the challenge of estimating annotator expertise in scenarios where the ground truth is

1118

1119

1120 unavailable. The key innovation of KAAR lies in its use of Centered Kernel
 1121 Alignment (CKA) to measure the similarity between input features and annotator
 1122 labels.

1123 Given a dataset with input features $\mathbf{X}$ and labels from multiple annotators $\mathbf{Y}$, KAAR
 1124 computes a kernel matrix $\mathbf{K}_\nu$ as a convex combination of R basis kernels:

$$\mathbf{K}_\nu = \sum_{r=1}^R \nu_r \mathbf{K}_r, \quad (3-1)$$

1125 where $\mathbf{K}_r$ is computed using a kernel function over the input features, and ν_r
 1126 represents the relevance weight for the r -th annotator. The weights are optimized
 1127 by maximizing the CKA between $\mathbf{K}_\nu$ and a target kernel $\mathbf{F}$ computed from the
 1128 input features:

$$\rho(\mathbf{K}_\nu, \mathbf{F}) = \frac{\langle \bar{\mathbf{K}}_\nu, \bar{\mathbf{F}} \rangle_{\text{F}}}{\|\bar{\mathbf{K}}_\nu\|_{\text{F}} \|\bar{\mathbf{F}}\|_{\text{F}}}, \quad (3-2)$$

1129 A key advantage of KAAR is its ability to capture dependencies between annotators
 1130 through the kernel framework, without requiring explicit parametric modeling of
 1131 these relationships.

1132 3.1.2 Localized Kernel Alignment-Based Annotator 1133 Relevance Analysis (LKAAR)

1134 LKAAR extends KAAR by introducing locality in the kernel alignment framework.
 1135 Rather than assuming constant annotator performance across the input space,

1136 LKAAR models the annotator relevance as a function of the input features. The
 1137 kernel combination in LKAAR takes the form:

$$K_q = \sum_{r=1}^R Q_r K_r Q_r, \quad (3-3)$$

1138 where Q_r is a diagonal matrix whose elements represent the local relevance of
 1139 annotator r . These elements are computed as:

$$q_r(\mathbf{x}_n) = \begin{cases} \beta_0^{(r)} + \sum_{n'=1}^N \beta_{n'}^{(r)} \kappa_\beta(\mathbf{x}_n, \mathbf{x}_{n'}), & \text{if } n \in N_r \\ 0, & \text{Otherwise} \end{cases} \quad (3-4)$$

1140 This localized approach allows LKAAR to better model inconsistent annotators
 1141 whose performance varies across different regions of the input space.

1142 **3.1.3 Regularized Chained Deep Neural Network** 1143 **(RCDNN)**

1144 RCDNN represents a deep learning approach to multiple annotator learning,
 1145 inspired by the chained Gaussian processes model. The method models both the
 1146 ground truth estimation and annotator reliability through a deep neural network
 1147 architecture. The likelihood function in RCDNN takes the form:

$$p(\mathbf{Y}|\boldsymbol{\theta}) = \prod_{n=1}^N \prod_{r \in R_n} \left(\prod_{k=1}^K \zeta_{n,k}^{\delta(y_n^{(r)} - k)} \right)^{\lambda_n^{(r)}} \left(\frac{1}{K} \right)^{1 - \lambda_n^{(r)}}, \quad (3-5)$$

1148 where $\lambda_n^{(r)}$ represents the reliability of annotator r for instance n , and $\zeta_{n,k}$ is the
 1149 estimated probability of class k for instance n .

1150 The RCDNN architecture is built upon a sophisticated multi-layer structure that
 1151 combines several key innovations in deep learning. At its core, it employs multiple
 1152 dense layers with carefully chosen activation functions, each designed to capture
 1153 different aspects of the annotator-data relationship. To prevent overfitting, a
 1154 comprehensive regularization strategy is implemented, combining both l1 and l2
 1155 norms with dropout mechanisms. The network architecture splits into specialized
 1156 branches: one dedicated to estimating the ground truth labels, and another
 1157 focused on modeling annotator reliability. This dual-branch approach allows the
 1158 network to simultaneously learn both the true underlying patterns in the data and
 1159 the individual characteristics of each annotator. Furthermore, the implementation
 1160 of Monte Carlo dropout during prediction provides a mechanism for uncertainty
 1161 estimation, offering not just predictions but also confidence measures for those
 1162 predictions.

1163 3.1.4 Chained Gaussian Processes Model

1164 The Chained Gaussian Processes (CGP) model provides a probabilistic framework for
 1165 modeling multiple outputs through a chain of Gaussian processes. In the context of
 1166 multiple annotators, CGP can be used to model both the ground truth and annotator
 1167 behavior. The key idea is to model a likelihood function with J parameters as in
 1168 [Giraldo et al., 2022]:

$$p(\mathbf{y}|\mathbf{X}, \boldsymbol{\theta}) = \prod_{n=1}^N p(y_n | \theta_1(\mathbf{x}_n), \dots, \theta_J(\mathbf{x}_n)), \quad (3-6)$$

1169 where each parameter $\theta_j(\mathbf{x})$ is modeled using a separate Gaussian process. This
 1170 framework provides remarkable flexibility in modeling complex relationships

1171 within the data. The use of Gaussian processes enables natural uncertainty
1172 quantification through posterior distributions, allowing the model to express
1173 varying degrees of confidence in its predictions. The framework inherently
1174 handles missing data through its probabilistic nature, and the automatic relevance
1175 determination property of GPs helps identify the most important features for
1176 prediction.

1177 The CGP framework becomes particularly powerful when integrated with concepts
1178 from KAAR and LKAAR. By incorporating kernel alignment principles, the model
1179 can make more informed choices about kernel functions, better capturing the
1180 underlying structure of the data. The framework can be extended to include
1181 input-dependent lengthscales, allowing for varying degrees of smoothness across
1182 the input space. Furthermore, through multi-output GP formulations, the model
1183 can capture complex dependencies between different annotators, providing a rich
1184 representation of how different experts' opinions relate to each other.

1185 However, implementing CGP in practice presents several significant challenges.
1186 The computational complexity of the model scales cubically with the dataset size,
1187 making it potentially prohibitive for large-scale applications. The need for
1188 approximate inference techniques in non-Gaussian likelihood scenarios can
1189 introduce additional complexity and potential approximation errors. When
1190 dealing with high-dimensional inputs, the model may struggle with the curse of
1191 dimensionality, requiring careful feature selection or dimensionality reduction
1192 preprocessing. Additionally, the selection of appropriate kernel functions remains
1193 a critical challenge, as it significantly impacts the model's performance and its
1194 ability to capture relevant patterns in the data.

1195 3.2 Tooling and Implementation

1196 The implementation of Chained Gaussian Processes for multiple annotator
1197 learning requires robust and flexible software tools that can handle both the

1198 probabilistic nature of Gaussian processes and the complexity of deep
1199 architectures. In our implementation, we leverage two key frameworks: GPflow
1200 and GPflux. These tools provide the foundation for building scalable and efficient
1201 implementations of our models while maintaining the theoretical rigor necessary
1202 for proper uncertainty quantification.

1203 The choice of these frameworks is motivated by several factors. First, both are
1204 built on top of TensorFlow, providing access to automatic differentiation and GPU
1205 acceleration capabilities. Second, they offer a modular design that allows for easy
1206 extension and modification, which is crucial when implementing novel model
1207 architectures. Third, they provide robust implementations of various Gaussian
1208 process models and inference methods, allowing us to focus on the specific
1209 challenges of multiple annotator learning rather than reimplementing basic GP
1210 functionality.

1211 3.2.1 GPflow Overview

1212 GPflow is a Python package for building Gaussian process models in TensorFlow.
1213 Developed as a modern and scalable alternative to GPy, it leverages TensorFlow’s
1214 computational capabilities to provide efficient implementation of Gaussian process
1215 models. The framework is designed with four key principles in mind: ease of use,
1216 computational efficiency, extensibility, and automatic differentiation. This work
1217 was originally published in [de G. Matthews et al., 2016].

1218 At its core, GPflow provides implementations of various Gaussian process models
1219 including GPR (Gaussian Process Regression), GPMC (Gaussian Process Monte
1220 Carlo), SGPR (Sparse Gaussian Process Regression), and SGPMC (Sparse Gaussian
1221 Process Monte Carlo). These implementations are built on top of TensorFlow’s
1222 computational graph framework, allowing for automatic differentiation and GPU
1223 acceleration where available.

1224 The framework’s architecture is modular, with clear separation between model
1225 components such as kernels, likelihoods, and optimization routines. This
1226 modularity makes it straightforward to extend existing models or create new ones.
1227 GPflow includes a comprehensive suite of kernel functions, ranging from the basic
1228 RBF (Radial Basis Function) kernel to more sophisticated ones like the Spectral
1229 Mixture kernel.

1230 One of GPflow’s distinguishing features is its robust handling of model parameters.
1231 It provides facilities for parameter transformation (ensuring parameters remain in
1232 valid ranges during optimization), prior specification, and flexible optimization
1233 strategies. The framework also includes utilities for model evaluation and
1234 prediction, with built-in support for uncertainty estimation.

1235 3.2.2 GPflux Framework

1236 GPflux extends GPflow to enable the construction and training of Deep Gaussian
1237 Process (DGP) models using modern deep learning practices. Built on top of both
1238 GPflow and TensorFlow, GPflux provides a flexible framework for implementing
1239 deep probabilistic models that combine the expressiveness of deep neural
1240 networks with the uncertainty quantification capabilities of Gaussian processes.
1241 This work was originally published in [Dutordoir et al., 2021].

1242 The framework implements Deep Gaussian Processes as a composition of Gaussian
1243 process layers, where each layer transforms its inputs through a Gaussian process
1244 mapping. This hierarchical structure allows the model to capture complex,
1245 non-stationary patterns in data while maintaining uncertainty estimates
1246 throughout the entire processing pipeline. GPflux handles this complexity through
1247 a carefully designed architecture that separates the concerns of model
1248 specification, inference, and prediction.

1249 A key innovation in GPflux is its integration with Keras-style layers, allowing
1250 practitioners to combine deterministic neural network layers with Gaussian

process layers in a single model. This hybrid approach enables the construction of models that can leverage both the structural assumptions encoded in neural network architectures and the flexibility of Gaussian processes.

For inference, GPflux implements various approximate methods, including stochastic variational inference with inducing points, which makes it possible to train deep Gaussian process models on large datasets. The framework provides built-in support for minibatch training, making it practical to work with datasets that don't fit in memory.

3.3 Experimental Setup

Synthetic datasets were used to evaluate the proposed methods in classification and regression tasks, following very similar procedures to the ones used in [Gil-Gonzalez et al., 2023], which are as follows:

3.3.1 Classification

A fully synthetic data for a 1-D ($P = 1$) multiclass classification problem with $K = 3$ classes is generated. The input feature matrix $\mathbf{X}$ is constructed by randomly sampling $N = 100$ points from a uniform distribution over the interval $[0, 1]$. For each sample n , the true label is determined by computing three target values:

$$\begin{aligned} t_{n,1} &= \sin(2\pi x_n), \\ t_{n,2} &= -\sin(2\pi x_n), \\ t_{n,3} &= -\sin(2\pi(x_n + 0.25)) + 0.5, \end{aligned} \tag{3-7}$$

and assigning the class label as the index $i \in \{1, 2, 3\}$ that maximizes $t_{n,i}$. This construction ensures a nontrivial, nonlinear decision boundary between the classes.

For evaluation, the test set is generated by extracting 200 equally spaced samples from the interval $[0, 1]$, providing a dense coverage of the input space to assess generalization performance.

Note that at this point, the fully synthetic dataset does not contain real annotator labels. Therefore, it is necessary to simulate annotator labels as corrupted versions of the hidden ground truth. In these experiments, the simulation is designed to account for: 1) dependencies among annotators, and 2) labeler performance that varies as a function of the input features.

To achieve this, an **SLFM**-based approach (termed SLFM-C) is employed, which generates annotator labels as follows:

1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q : \mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R$, $n \in N$.
2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$, where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D representation of the input features in $\mathbf{X}$, obtained using the t-distributed Stochastic Neighbor Embedding (t-SNE) approach.
3. **Reliability calculation:** Calculate $\hat{\lambda}_n^r = \varsigma(\hat{f}_{l,r,n})$, where $\varsigma(\cdot) \in [0, 1]$ is the sigmoid function.
4. **Label assignment:** The r th annotator's label for the n th sample is given by

$$y_n^r = \begin{cases} y_n, & \text{if } \hat{\lambda}_n^r \geq 0.5 \\ \tilde{y}_n, & \text{if } \hat{\lambda}_n^r < 0.5 \end{cases}$$

where y_n is the true label and $\tilde{y}_n$ is a flipped version of the actual label y_n .

This procedure ensures that the simulated annotator labels exhibit both inter-annotator dependencies and input-dependent reliability.

1292 3.3.2 Regression

1293 A fully synthetic dataset is also generated for a 1-D regression problem, where the
1294 ground truth for the n th sample is defined as:

$$y_n = \sin(2\pi x_n) \sin(6\pi x_n) \quad (3-8)$$

1295 Here, the input matrix $\mathbf{X}$ is constructed by randomly sampling 100 points uniformly
1296 from the interval $[0, 1]$. The test set is created by extracting equally spaced samples
1297 from the same interval, ensuring a clear separation between training and testing
1298 data.

1299 To simulate the presence of multiple annotators, the ground truth is corrupted with
1300 annotator-specific noise. For each annotator r , the observed label for the n th sample
1301 is given by:

$$y_n^r = y_n + \epsilon_n^r$$

1302 where $\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$ is Gaussian noise, and v_n^r represents the error variance for
1303 the r th annotator, which may depend on the input features. This setup allows us
1304 to model both the bias and the reliability of each annotator, as well as potential
1305 dependencies between annotators.

1306 To further increase realism, the error variance v_n^r can be modeled as a function of
1307 the input features, capturing scenarios where annotator reliability varies across the
1308 input space, thus:

$$\epsilon_n^r \sim \mathcal{N}(0, v_n^r)$$

1309 being v_n^r , the r th annotator's error variance for the n th sample.

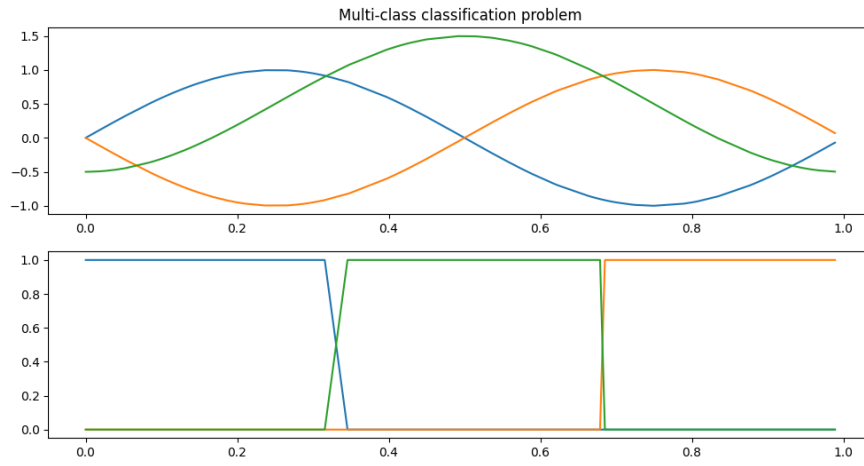
1310 Again, for modeling the error as a function of the input features, an **SLFM**-based
1311 approach (termed SLFM-R) is employed to build the noisy labels, as follows:

- 1312 1. **Definition of deterministic functions:** Define Q deterministic functions $\hat{\mu}_q :$
1313 $\mathcal{X} \rightarrow \mathbb{R}$ and their combination parameters $\hat{w}_{l,r,q} \in \mathbb{R}$, for all $r \in R, n \in N$.
- 1314 2. **Annotator performance computation:** Compute $\hat{f}_{l,r,n} = \sum_{q=1}^Q \hat{w}_{l,r,q} \hat{\mu}_q(\hat{x}_n)$,
1315 where $\hat{x}_n \in \mathbb{R}$ is the n th component of $\hat{\mathbf{x}} \in \mathbb{R}^N$. Here, $\hat{\mathbf{x}}$ is the 1-D
1316 representation of the input features in $\mathbf{X}$, obtained using the t-distributed
1317 Stochastic Neighbor Embedding (t-SNE) approach.
- 1318 3. **Compute variance :** Finally, determine $v_n^r = \exp(\hat{f}_{l,r,n})$.

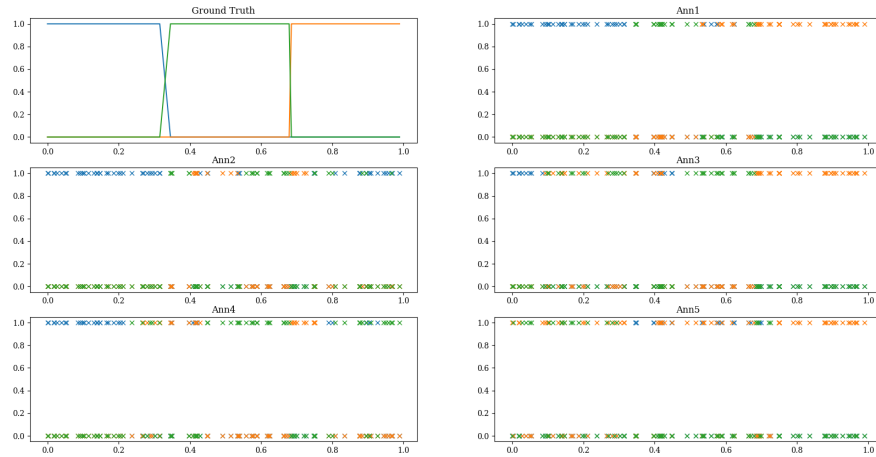
1319 3.4 Results

1320 The built synthetic datasets are used to train the **CCGPMA** model. Emulated
1321 classification labels following the true label determining function from 3-7 can be
1322 seen in Figure 3-1. On the other hand, the ground truth and noisy labels emulation
1323 for annotators across the feature space for regression task can be seen in Figure
1324 3-3.

1325 Using both GPflux and GPflow for the regression and classifications tasks, leads to
1326 the following combinations experiments: GPflux-CCGPMA Regression,
1327 GPflow-CCGPMA Regression, GPflux-CCGPMA Binary Classification,
1328 GPflow-CCGPMA Binary Classification, GPflow-CCGPMA Multiclass Classification,
1329 GPflux-CCGPMA Multiclass Classification. All of the experiments converged
1330 successfully with almost identical results and close to no difference in
1331 performance.



(a) Ground truth labels across the feature space



(b) Noisy emulations for annotators across the feature space for classification task

Figure 3-1 Synthetic dataset overview for classification task

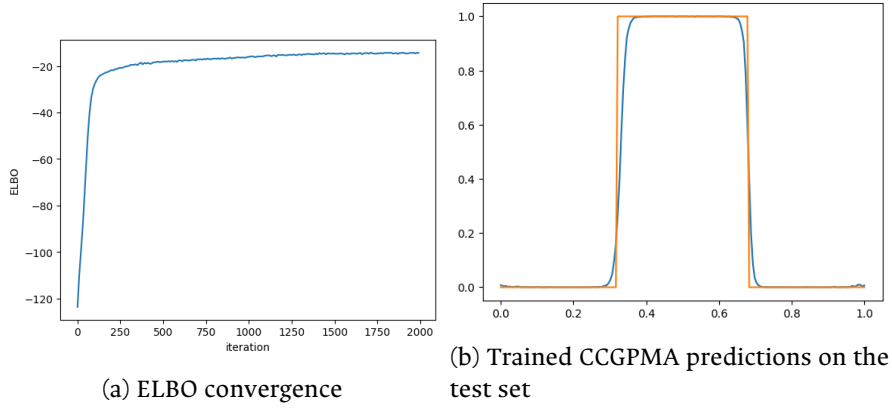


Figure 3-2 Classification results for the CCGPMA model using GPflux.

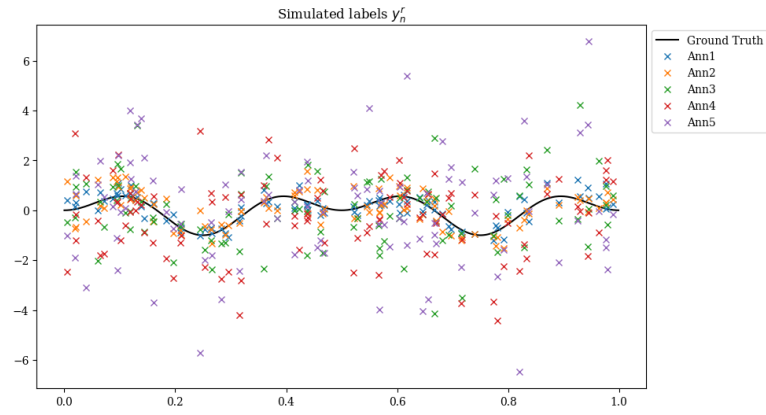
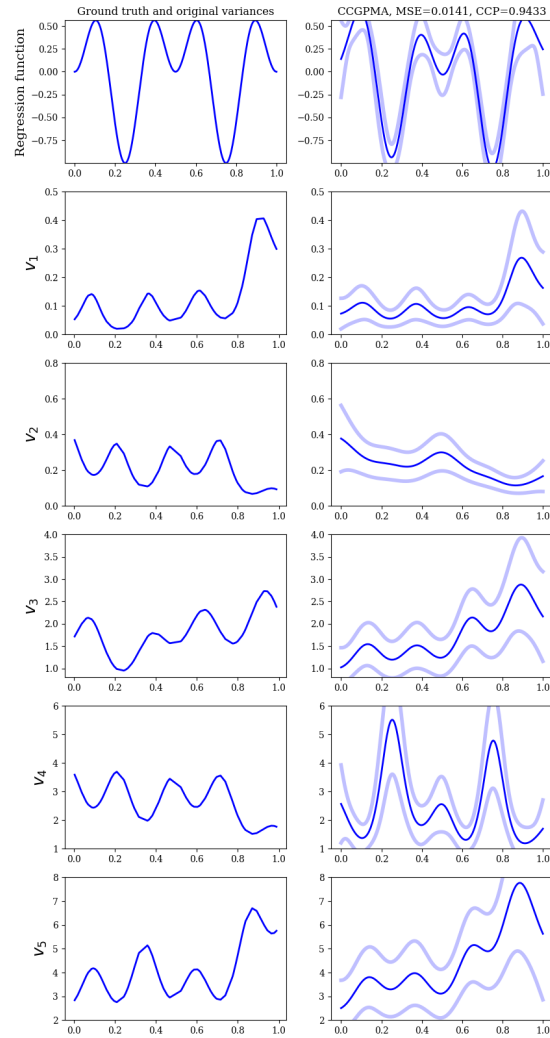


Figure 3-3 Ground truth and noisy labels emulation for annotators across the feature space for regression task.



(a) Trained CCGPMA predictions on the test set, for inference of both the hidden ground truth and the annotators' reliabilities across the feature space.

Figure 3-4 Regression results for the CCGPMA model using GPflux.

3.5 Conclusion

3.5.1 Comparative Analysis

The evolution of multiple annotator learning methods represents a fascinating progression in how we approach the challenge of learning from diverse expert opinions. KAAR established a foundation with its non-parametric approach through kernel alignment, though it operates under the assumption of constant annotator performance across the input space. LKAAR advanced this concept by introducing input-dependent annotator behavior, recognizing that expert performance often varies across different types of inputs. RCDNN brought the power and flexibility of deep learning to the problem, offering excellent scalability but requiring careful attention to regularization and architecture design. CGP represents perhaps the most theoretically elegant approach, providing a principled probabilistic framework, though it faces significant computational challenges in practical applications.

The selection of an appropriate method for a given application requires careful consideration of several factors. The size and dimensionality of the dataset play a crucial role, as some methods scale better than others with increasing data volume. The need for uncertainty quantification may favor probabilistic approaches like CGP, while computational constraints might push towards more efficient methods like KAAR. The presence of input-dependent annotator behavior could make LKAAR or RCDNN more appropriate choices. Additionally, when interpretability is a key requirement, as in many medical applications, methods like KAAR and LKAAR might be preferred over the more complex deep learning approaches. Table 3-1 provides a comprehensive comparison of the key characteristics and capabilities of each method discussed.

Table 3-1 Comparison of Multiple Annotator Learning Methods

Characteristic	KAAR [Gil-Gonzalez et al., 2018]	LKAAR [Gil-Gonzalez et al., 2021a]	RCDNN [Gil-Gonzalez et al., 2021b]	CGP [Julían and Álvarez Meza Andrés Marino, 2023]
Input-dependent annotator behavior	No	Yes	Yes	Yes
Annotator dependencies modeling	Yes	Yes	Yes	Yes
Uncertainty quantification	No	No	Partial	Yes
Computational scalability	High	Medium	High	Low
Missing data handling	Yes	Yes	Yes	Yes
Non-linear relationships	Yes	Yes	Yes	Yes
Training complexity	Low	Medium	High	High
Hyperparameter sensitivity	Low	Medium	High	Medium
Interpretability	High	Medium	Low	Medium
Memory requirements	Low	Medium	High	High

Note: Partial uncertainty quantification in RCDNN refers to the use of Monte Carlo dropout for approximate uncertainty estimation.

3.5.2 Implementation Considerations

When implementing Chained Gaussian Processes for multiple annotator learning using these tools, several key considerations must be addressed. First, the model architecture must be carefully designed to capture both the individual annotator characteristics and their interdependencies. This requires extending the basic GP layers provided by GPflux to handle multiple output streams - one for each annotator's reliability model.

Second, the inference procedure must be adapted to handle the unique challenges of multiple annotator data, such as missing labels and varying levels of annotator expertise across different regions of the input space. The variational inference capabilities of GPflux provide a solid foundation for this, but careful attention must be paid to the design of the variational approximation to ensure it captures the relevant uncertainty in both the ground truth and annotator reliability estimates.

Third, kernel selection and design presents a critical challenge in implementing the CGP model. The choice of kernel functions significantly impacts the model's ability to capture the underlying structure of the data and the relationships between annotators. Several considerations must be taken into account:

- **Kernel Complexity:** The model requires multiple kernel functions to capture different aspects of the data - one for the ground truth and additional kernels for each annotator's reliability model. This multi-kernel setup increases the complexity of kernel selection and optimization.

- 1379 • **Input-Dependent Behavior:** Since annotator performance often varies across
 1380 the input space, the kernel functions must be flexible enough to capture this
 1381 non-stationary behavior. This may require using composite kernels or kernels
 1382 with input-dependent lengthscales.
- 1383 • **Inter-Annotator Dependencies:** The kernel design must account for potential
 1384 correlations between different annotators' behaviors. This can be achieved
 1385 through careful design of the kernel structure in the SLFM framework, where
 1386 the combination of basis kernels must be chosen to effectively model these
 1387 dependencies.
- 1388 • **Computational Efficiency:** The choice of kernel functions also impacts
 1389 computational performance. While more expressive kernels may better
 1390 capture complex patterns, they often come with increased computational
 1391 cost. This trade-off must be carefully considered, especially when dealing
 1392 with large datasets.

1393 Finally, computational efficiency becomes a critical concern when scaling to large
 1394 datasets with multiple annotators. The minibatch training capabilities of both
 1395 frameworks help address this, but additional considerations such as appropriate
 1396 inducing point selection and batch size tuning become important for achieving
 1397 good performance in practice.

1398 3.5.3 Final Remarks

1399 In this chapter, we have presented a comprehensive overview of the Chained
 1400 Gaussian Processes model and its application to multiple annotator learning. We
 1401 have discussed the theoretical underpinnings of the model, its implementation
 1402 using GPflow and GPflux, and its evaluation on synthetic datasets. It was shown in
 1403 [Gil-Gonzalez et al., 2023] that CCGPMA outperforms other method in state of the
 1404 art methods in terms of accuracy and uncertainty quantification (see 1.2.4).

1405 While Gaussian Processes have demonstrated remarkable success in regression
1406 and classification tasks, their application to image segmentation presents
1407 significant computational challenges. The cubic computational complexity of GPs
1408 ($\mathcal{O}(N^3)$), which scales with the number of data points, becomes particularly
1409 problematic when dealing with high-resolution images where each pixel can be
1410 considered a data point. This limitation makes traditional GP approaches
1411 impractical for large-scale image segmentation tasks, where the number of pixels
1412 can easily reach into the millions. Alternative approaches, such as deep
1413 learning-based segmentation methods, often provide more computationally
1414 efficient solutions for these high-dimensional problems, though at the cost of
1415 uncertainty quantification capabilities that GPs naturally provide.

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CHAPTER

FOUR

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TRUNCATED GENERALIZED CROSS ENTROPY FOR SEGMENTATION

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4.1 Loss Functions for Multiple Annotators

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As mentioned in Section 2.3.2, a loss function is a key element for defining the objective function of a deep learning model. The categorical cross-entropy loss is a common loss function for classification tasks. However, in the case of multiple annotators, the categorical cross-entropy loss is not able to handle the varying reliability of the annotators. In this section, we will propose a loss function that is able to handle multiple annotators' segmentation masks while accounting for their varying reliability across different regions of the image.

1430 4.1.1 Generalized Cross Entropy

1431 The Generalized Cross Entropy (GCE) loss function was first introduced by [Zhang
1432 and Sabuncu, 2018] as a robust alternative to the standard cross-entropy loss,
1433 particularly effective in handling noisy labels. Let us first consider the Cross
1434 Entropy (CE) and Mean Absolute Error (MAE) loss functions:

$$MAE(\mathbf{y}, f(\mathbf{x})) = \|\mathbf{y} - f(\mathbf{x})\|_1 \quad (4-1)$$

$$CE(\mathbf{y}, f(\mathbf{x})) = \sum_{k=1}^K y_k \log(f_k(\mathbf{x})) \quad (4-2)$$

1435 where $y_k \in \mathbf{y}$, $f_k(\mathbf{x}) \in f(\mathbf{x})$, and $\|\cdot\|_1$ stands for the l_1 -norm. Of note, $\mathbf{1}^\top \mathbf{y} =$
1436 $\mathbf{1}^\top f(\mathbf{x}) = 1$, $\mathbf{1} \in \{1\}^K$ being an all-ones vector. In addition, the MAE loss can be
1437 rewritten for softmax outputs, yielding:

$$MAE(\mathbf{y}, f(\mathbf{x})) = 2(1 - \mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x}))) \quad (4-3)$$

1438 where $\odot$ stands for the element-wise product.

1439 The CE loss function exhibits several distinct characteristics that make it
1440 particularly sensitive to noisy labels. It is unbounded from above and heavily
1441 penalizes confident but wrong predictions, which can lead to instability in the
1442 presence of label noise. In contrast, the MAE loss offers a more robust alternative
1443 with its bounded nature and symmetric properties in softmax-based
1444 representations. While the MAE loss is more resilient to noisy labels, it tends to

1445 train more slowly due to its equal weighting of all mistakes regardless of
1446 confidence.

1447 The GCE loss function, as defined by the authors in [Zhang and Sabuncu, 2018],
1448 provides a flexible framework that bridges these two approaches:

$$GCE(\mathbf{y}, f(\mathbf{x})) = 2 \frac{1 - (\mathbf{1}^\top (\mathbf{y} \odot f(\mathbf{x})))^q}{q}, \quad (4-4)$$

1449 with $q \in (0, 1]$. Remarkably, the limiting case for $q \rightarrow 0$ in GCE is equivalent to the
1450 CE expression, and when $q = 1$, it equals the MAE loss. In addition, the GCE holds
1451 the following gradient with regard to θ :

$$\frac{\partial GCE(\mathbf{y}, f(\mathbf{x}; \theta) | k)}{\partial \theta} = -f_k(\mathbf{x}; \theta)^{q-1} \nabla_\theta f_k(\mathbf{x}; \theta). \quad (4-5)$$

1452 The GCE loss combines the best of both worlds, offering robustness to label noise
1453 while maintaining the convexity property necessary for optimization. The
1454 truncation parameter q provides a mechanism to control the sensitivity to outliers,
1455 allowing for fine-tuning of the loss function's behavior based on the specific
1456 characteristics of the dataset.

1457 4.1.2 Extension to Multiple Annotators

1458 In the context of multiple annotators, we need to consider the varying reliability
1459 of each annotator across different regions of the image. Let's consider a k -class
1460 multiple annotators segmentation problem with the following data representation:

$$\mathbf{X} \in \mathbb{R}^{W \times H}, \{\mathbf{Y}_r \in \{0, 1\}^{W \times H \times K}\}_{r=1}^R; \quad \mathbf{\Psi} \in [0, 1]^{W \times H \times K} = f(\mathbf{X}) \quad (4-6)$$

1461 where the segmentation mask function maps the input to output as:

$$f : \mathbb{R}^{W \times H} \rightarrow [0, 1]^{W \times H \times K} \quad (4-7)$$

1462 The segmentation masks $\mathbf{Y}_r$ satisfy the following condition for being a softmax-like
1463 representation:

$$\mathbf{Y}_r[w, h, :] \mathbf{1}_k^\top = 1; \quad w \in W, h \in H \quad (4-8)$$

1464 4.1.3 Reliability Maps and Truncated GCE

1465 The key innovation in our approach is the introduction of reliability maps Λ_r for
1466 each annotator:

$$\left\{ \Lambda_r(\mathbf{X}; \theta) \in [0, 1]^{W \times H} \right\}_{r=1}^R \quad (4-9)$$

1467 These reliability maps serve as a sophisticated mechanism to estimate the
1468 confidence of each annotator at every spatial location (w, h) in the image. By
1469 learning these maps jointly with the segmentation model, the network gains the
1470 ability to adapt to the varying levels of expertise across different regions of the
1471 image. This approach allows for dynamic weighting of each annotator's

1472 contribution based on their estimated reliability in specific areas, effectively
 1473 handling cases where annotators might demonstrate different levels of expertise in
 1474 different parts of the image.

1475 The proposed Truncated Generalized Cross Entropy for Semantic Segmentation
 1476 (TGCE_{SS}) combines the robustness of GCE with the flexibility of reliability maps:

$$TGCE_{SS}(\mathbf{Y}_r, f(\mathbf{X}; \theta) | \Lambda_r(\mathbf{X}; \theta)) = \mathbb{E}_r \left\{ \mathbb{E}_{w,h} \left\{ \Lambda_r(\mathbf{X}; \theta) \circ \mathbb{E}_k \left\{ \mathbf{Y}_r \circ \left(\frac{\mathbf{1}_{W \times H \times K} - f(\mathbf{X}; \theta)^{\circ q}}{q} \right); k \in K \right\} + \right. \right. \\ \left. \left. (\mathbf{1}_{W \times H} - \Lambda_r(\mathbf{X}; \theta)) \circ \left(\frac{\mathbf{1}_{W \times H} - (\frac{1}{K} \mathbf{1}_{W \times H})^{\circ q}}{q} \right); w \in W, h \in H \right\}; r \in R \right\} \quad (4-10)$$

1477 where $q \in (0, 1)$ controls the MAE or CE level in the same way as in the GCE loss
 1478 function. The loss function consists of two main components:

- 1479 • The first term weighted by Λ_r represents the GCE loss for regions where the
 1480 annotator is considered reliable
- 1481 • The second term weighted by $(1 - \Lambda_r)$ provides a uniform prior for regions
 1482 where the annotator is considered unreliable

1483 For a batch containing N samples, the total loss is computed as:

$$\mathcal{L}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) = \frac{1}{N} \sum_n^N TGCE_{SS}(\mathbf{Y}_r[n], f(\mathbf{X}[n]; \theta) | \Lambda_r(\mathbf{X}[n]; \theta)) \quad (4-11)$$

1484 4.1.4 Discussion and Conclusions

1485 This chapter presents a novel approach to handle multiple annotators in semantic
 1486 segmentation tasks through the introduction of the Truncated Generalized Cross
 1487 Entropy for Semantic Segmentation (TGCE_{SS}) loss function. The key contributions
 1488 and novelties of this work can be summarized as follows:

- 1489 • **Reliability-aware Learning:** Unlike traditional approaches that treat all
 1490 annotators equally, the proposed method introduces reliability maps (Λ_r)
 1491 that dynamically estimate the confidence of each annotator at every spatial
 1492 location in the image. This allows the model to adapt to varying levels of
 1493 expertise across different regions.
- 1494 • **Robust Loss Function:** The proposed TGCE_{SS} combines the benefits of both
 1495 Cross Entropy and Mean Absolute Error losses through the Generalized
 1496 Cross Entropy framework. The truncation parameter q provides a flexible
 1497 mechanism to control the sensitivity to noisy labels while maintaining the
 1498 convexity property necessary for optimization.
- 1499 • **Unified Framework:** The approach provides a unified framework that
 1500 handles both reliable and unreliable regions through a weighted
 1501 combination of GCE loss and uniform prior. This is particularly important in
 1502 medical imaging where annotator expertise can vary significantly across
 1503 different anatomical structures.

1504 The main novelty of this approach lies in its ability to simultaneously learn the
 1505 segmentation task while estimating annotator reliability in a spatially-aware
 1506 manner. This represents a significant advancement over existing methods that
 1507 either:

- 1508 • Treat all annotators equally without considering their varying expertise
- 1509 • Use global reliability scores that don't account for spatial variations in
 1510 annotator performance

- Rely on fixed, pre-defined reliability measures rather than learning them from data

The proposed $TGCE_{SS}$ loss function represents a step forward in handling the complex nature of multiple annotator scenarios in semantic segmentation, particularly in medical imaging applications where expert knowledge can vary significantly across different regions of interest. This work provides a foundation for more sophisticated approaches to handling multiple annotators in deep learning-based segmentation tasks.

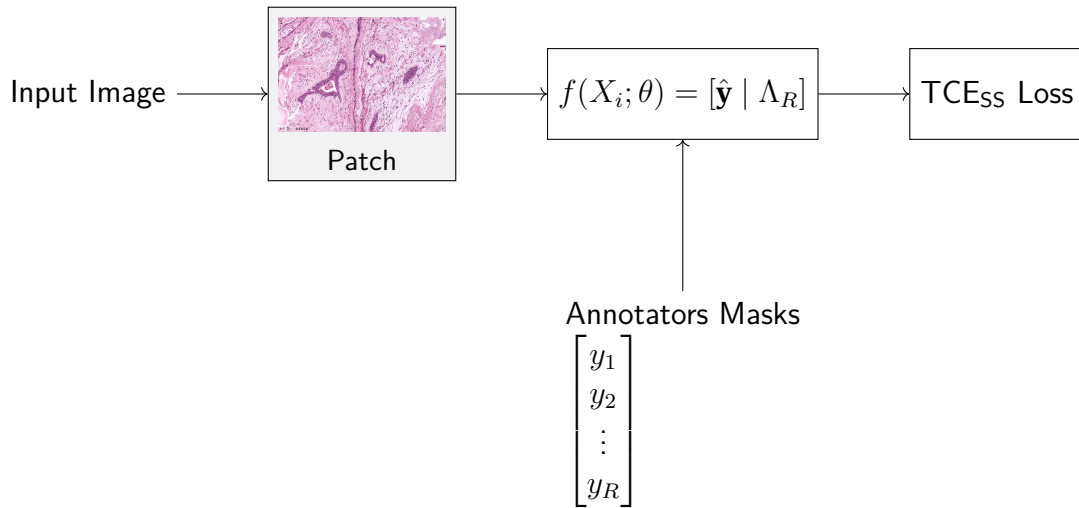


Figure 4-1 Working mechanism of the proposed Truncated Generalized Cross Entropy for Semantic Segmentation ($TGCE_{SS}$) loss function. The loss combines reliability maps Λ_r with the model's predictions $f(\mathbf{X}; \theta)$ to compute a weighted loss that accounts for annotator reliability across different image regions.

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CHAPTER

FIVE

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CHAINED DEEP LEARNING AND $TGCE_{SS}$ LOSS FOR IMAGE SEGMENTATION

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As mentioned in Chapter 1, U-shaped CNNs have been proven a good solution for segmentation tasks in medical images, due to their ability to capture both global and local information with relatively low datasets.

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The proposed model architecture combines the strengths of U-Net with a ResNet-34 backbone, specifically designed to work with the $TGCE_{SS}$ loss function by providing a multi-output network which focuses on two tasks: **segmentation** and **reliability estimation**. The architecture is illustrated in Figure 5-2.

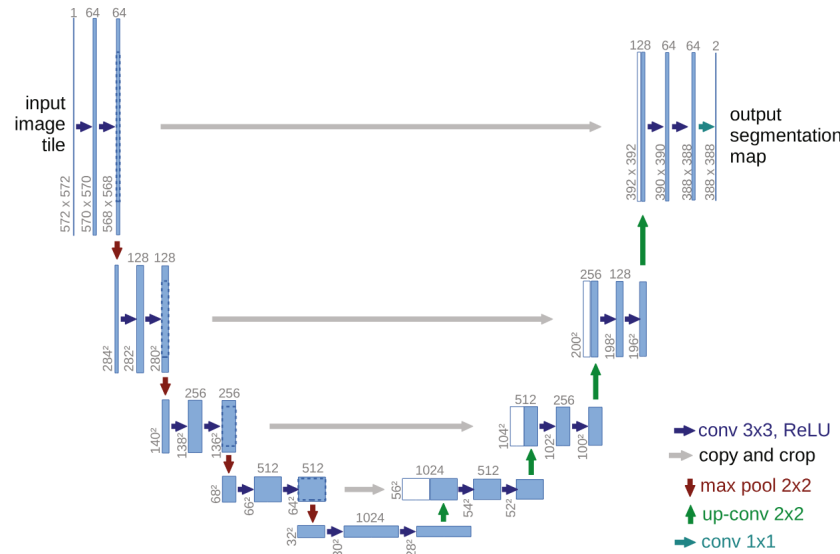


Figure 5-1 Original U-NET architecture.

5.0.1 Backbone Architecture

The model employs a pre-trained ResNet-34 as its encoder backbone, leveraging its deep residual learning framework for efficient feature extraction. The choice of ResNet-34 provides several key advantages: efficient feature extraction through residual connections, pre-trained weights that capture rich visual representations, and stable gradient flow during training. The ResNet-34 backbone is modified to serve as the encoder in our UNET architecture by removing the final fully connected layer and utilizing the feature maps from different stages of the network for skip connections.

5.0.2 U-Net Architecture

The U-Net architecture follows a traditional encoder-decoder structure with skip connections, where the encoder path implements the ResNet-34 structure. The decoder path employs transposed convolutions for upsampling, creating a

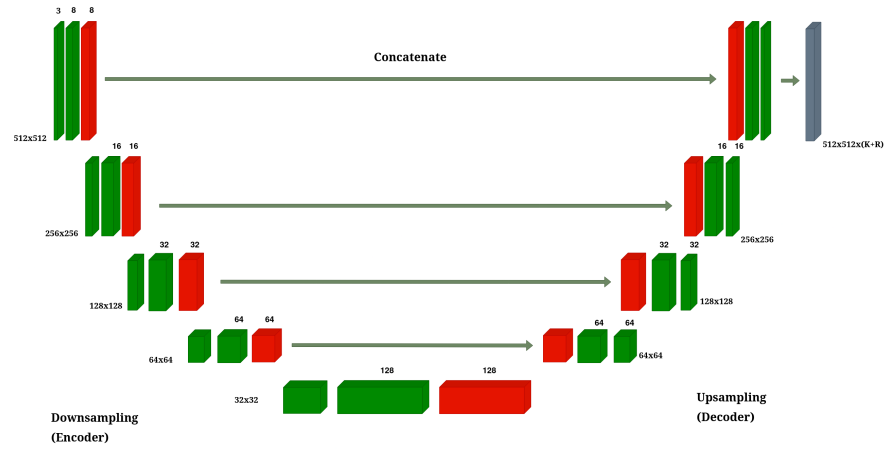


Figure 5-2 Overview of the proposed model architecture.

symmetrical architecture that effectively captures both high-level and low-level features. The architecture incorporates four downsampling stages in the encoder, corresponding to the ResNet-34 blocks, and four upsampling stages in the decoder. These stages are connected through skip connections that bridge corresponding encoder and decoder stages, allowing the network to preserve fine-grained details. Each convolution operation is followed by batch normalization and ReLU activation to ensure stable training and effective feature learning.

This U-Net architecture is expected to be coupled with a reliability branch which leads to a multi-output network. Three different approaches are considered for this branch: **scalar**, **features**, and **pixel**.

5.0.3 Scalar Reliability Estimation

The scalar reliability estimation branch is the simplest one, in which the reliability branch is a parallel branch to the segmentation branch, which leads to a multi-output network. The final output consists of R reliability maps Λ_r , one for each annotator, with values constrained to the $[0, 1]$ range through a sigmoid activation function. This design allows the network to learn and adapt to the varying reliability of different annotators across different regions of the image.

The scalar reliability branch is implemented by taking the bottleneck features from the encoder and applying a global average pooling operation followed by a dense layer with sigmoid activation. This results in a single reliability value per annotator for each input image, making it computationally efficient but less spatially aware of reliability variations within the image.

5.0.4 Feature-based Reliability Estimation

The feature-based reliability estimation approach provides a more detailed spatial understanding of annotator reliability by operating on the bottleneck features of the encoder. This approach maintains the spatial information from the encoder's bottleneck while being more computationally efficient than pixel-wise reliability estimation.

The feature-based reliability branch is implemented by applying two convolutional layers with batch normalization to the bottleneck features:

- A first convolutional layer with 32 filters to process the bottleneck features
- A second convolutional layer that produces R feature maps, one for each annotator

This approach provides a coarse-grained spatial reliability map that captures regional variations in annotator reliability while maintaining computational efficiency. The resulting reliability maps have the same spatial dimensions as the bottleneck features, providing a middle ground between scalar and pixel-wise reliability estimation.

1583 5.0.5 Pixel-wise Reliability Estimation

1584 The pixel-wise reliability estimation approach provides the most detailed spatial
1585 understanding of annotator reliability by operating on the full-resolution features
1586 from the decoder. This approach allows for fine-grained reliability estimation at the
1587 pixel level, capturing local variations in annotator reliability.

1588 The pixel-wise reliability branch is implemented by applying three convolutional
1589 layers with batch normalization to the decoder’s output features:

- 1590 • Two initial convolutional layers with 32 filters each to process the features
- 1591 • A final convolutional layer that produces R reliability maps, one for each
- 1592 annotator

1593 This approach provides the most detailed spatial reliability information, allowing
1594 the model to capture pixel-level variations in annotator reliability. The resulting
1595 reliability maps have the same spatial dimensions as the input image, providing the
1596 finest granularity of reliability estimation among the three approaches.

1597 5.0.6 Integration with TGCE_{SS} Loss

1598 The model produces two distinct outputs: segmentation masks $\hat{\mathbf{Y}} = f(\mathbf{X}; \theta)$ and
1599 reliability maps $\{\Lambda_r(\mathbf{X}; \theta)\}_{r=1}^R$. These outputs work in tandem with the TGCE_{SS}
1600 loss function described in Section 4.1. The loss function simultaneously guides the
1601 learning of both the segmentation masks and reliability maps, ensuring that the
1602 model learns to balance the contributions of different annotators based on their
1603 estimated reliability.

1604 The TGCE_{SS} loss function is implemented differently for each reliability approach,
1605 with specific adaptations to handle the different spatial dimensions of the reliability
1606 maps. All implementations share common hyperparameters:

- 1607 • $q = 0.1$: Controls the truncation of the loss function
- 1608 • $\epsilon = 10^{-8}$: Small constant to prevent numerical instability
- 1609 • $\alpha_{reg} = 0.1$: Weight for the regularization term
- 1610 • $\alpha_{entropy} = 0.1$: Weight for the entropy regularization
- 1611 • $\alpha_{sum} = 0.1$: Weight for the sum-to-one constraint

1612 For the scalar reliability approach, the loss function operates on global reliability
 1613 values per annotator. The reliability maps are expanded to match the spatial
 1614 dimensions of the predictions through tiling operations. This approach is
 1615 computationally efficient but provides the same reliability value across all spatial
 1616 locations.

1617 The feature-based reliability approach requires additional processing to handle the
 1618 spatial dimensions of the bottleneck features. The reliability maps are resized
 1619 using bilinear interpolation to match the spatial dimensions of the predictions.
 1620 This approach provides coarse-grained spatial reliability information while
 1621 maintaining computational efficiency.

1622 The pixel-wise reliability approach operates directly on full-resolution reliability
 1623 maps, requiring no spatial transformations. This approach provides the most
 1624 detailed spatial reliability information but requires more computational resources.

1625 All three implementations include three regularization terms to ensure proper
 1626 learning of the reliability maps:

- 1627 • **Center Regularization:** $\alpha_{reg} \cdot \mathbb{E}[(\Lambda_r - 0.5)^2]$
- 1628 • **Entropy Regularization:** $-\alpha_{entropy} \cdot \mathbb{E}[\Lambda_r \log(1 + \Lambda_r) + (1 - \Lambda_r) \log(1 + (1 - \Lambda_r))]$
- 1629 • **Sum-to-One Regularization:** $\alpha_{sum} \cdot \mathbb{E}[(\sum_r \Lambda_r - 1)^2]$

1630 These regularization terms help prevent the model from assigning extreme
 1631 reliability values and ensure a balanced distribution of reliability across
 1632 annotators. The loss function also includes a noise tolerance parameter ($\eta = 0.1$)
 1633 to handle cases where the model's predictions differ from the annotators' labels.

1634 The final loss function combines the main $TGCE_{SS}$ terms with the regularization
 1635 terms:

$$\mathcal{L} = \mathbb{E}[\text{term}_1 + \text{term}_2] + \alpha_{reg} + \alpha_{entropy} + \alpha_{sum} \quad (5-1)$$

1636 where term_1 and term_2 are the main $TGCE_{SS}$ terms that handle the interaction
 1637 between predictions and reliability maps.

1638 5.0.7 Training Process

1639 The training process begins with the initialization of the ResNet-34 backbone
 1640 using pre-trained weights, providing a strong foundation for feature extraction.
 1641 The model's hyperparameters are optimized using Bayesian optimization through
 1642 Keras Tuner, which searches for optimal values of:

- 1643 • Learning rate: Sampled logarithmically between 10^{-5} and 10^{-2}
- 1644 • $TGCE_{SS}$ parameters:
 - 1645 – q : Controls truncation of the loss function (0.1 to 0.9)
 - 1646 – Noise tolerance: Controls model's robustness to label noise (0.1 to 0.9)
 - 1647 – Regularization weights: Control the balance between different loss
 - 1648 terms (0.01 to 0.5)

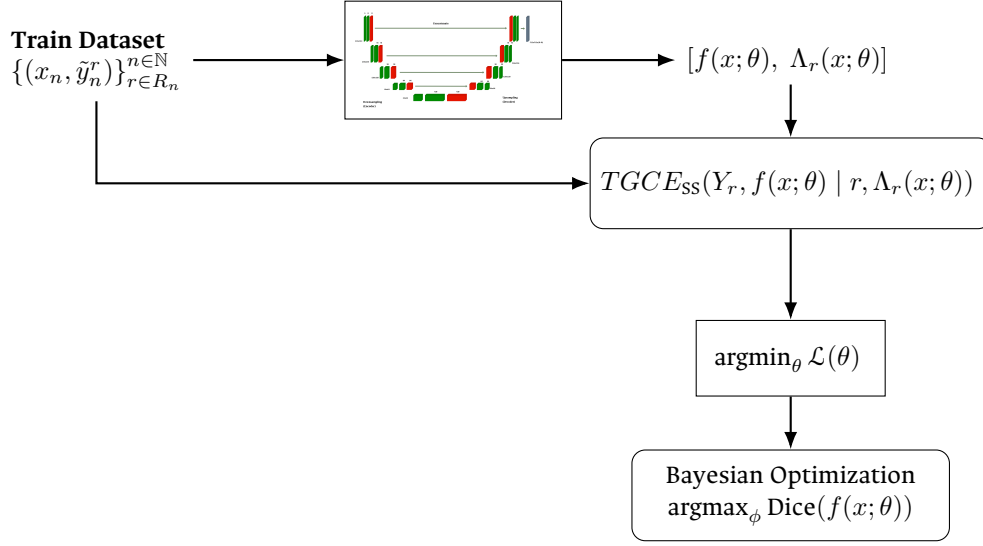


Figure 5-3 Training process of the proposed model overview. The model first undergoes hyperparameter optimization using Bayesian optimization, followed by end-to-end training with the optimal parameters. Estimated segmentation and reliability maps are computed for each input image, and the loss function is computed for the entire batch. Network parameters are tuned based on the loss function as objective (with minimization target) whereas the hyperparameters ϕ tuning are based on the Dice coefficient of the validation data set (with maximization target).

1649 The hyperparameter search is performed using a validation set, with the objective
 1650 of maximizing the Dice coefficient on the validation data. After finding the optimal
 1651 hyperparameters, the model is trained end-to-end using the Adam optimizer with
 1652 the best learning rate found. The $TGCE_{SS}$ loss function plays a crucial role in
 1653 updating both the segmentation and reliability branches, ensuring that the model
 1654 learns to effectively handle multiple annotators' inputs while accounting for their
 1655 varying reliability.

1656 The model's architecture is specifically designed to address the challenges of
 1657 multi-annotator segmentation. Through the ResNet-34 backbone, it learns robust
 1658 segmentation features that capture high-level patterns in the data. The UNET's
 1659 skip connections enable the preservation of fine-grained details, while the parallel
 1660 reliability branch allows the model to adapt to annotator-specific characteristics.
 1661 This comprehensive design enables the model to effectively handle multiple

1662 annotators' inputs while maintaining high segmentation accuracy and reliability
1663 estimation.

1664 5.1 Experimental Setup

1665 5.1.1 Noisy Annotations Generation

1666 A challenge arises for the creation of emulated noisy annotations datasets, since it
1667 is expected for images annotations to have some degree of expertise variability,
1668 similar to what is expected in real crowdsourced datasets. Simply introducing
1669 random noise into the annotated masks does not work, since the original
1670 morphological structures from the expected ground truth are far from being
1671 preserved. Instead, a “noisy” labeler is expected to produce an annotation which
1672 has at least some degree of morphological consistency on itself, even if it shows
1673 discrepancies when compared with some metric (like DICE score) against the
1674 ground truth annotation.

1675 This has been proven experimentally when introducing random noise into any
1676 popular segmentation dataset, in which the resulting mask is just a non coherent
1677 map of noise across the image, as shown in Figure 5-4.

1678 For this reason, a more sophisticated approach was needed to create datasets with
1679 emulated noisy annotations. The approach used here was to use a pre-trained
1680 U-Net model to produce a good enough ¹ segmentation of the image, which was
1681 then used to produce a “noisy” annotation by introducing random noise into the
1682 last encoder layers of the model, thus preserving the morphological consistency of
1683 the original annotation. This strategy works since slight modifications introduced

¹More precisely, a U-Net architecture with Resnet-34 backbone model with a simple categorical cross-entropy loss which reached 0.8426 Dice Coefficient after 44 epochs.

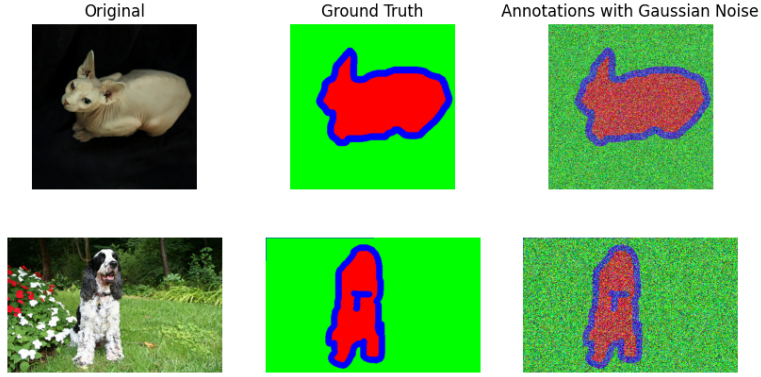


Figure 5-4 Example of a noisy mask generated by naively introducing random noise into a ground truth mask. Morphological consistency is lost.

1684 into the encoder layers weights, somehow resemble conceptual disturbances with
 1685 respect to the respect to the original ground truth label, which kind of emulates
 1686 the cultural behavior of having a different interpretation of the image, either for
 1687 having a different level of expertise or for having a different point of view or
 1688 school of thought. The first encoding layers were not modified, since it is expected
 1689 human labelers would agree on the most fundamental structures (analog to
 1690 extracted features from initial convolutional layers) in a similar way, thus
 1691 preserving the morphological consistency of the original annotation.

1692 In this way, the level of “disturbance” with respect to the ground truth for an
 1693 emulated annotator can be controlled by the level of noise introduced into the
 1694 weights of the last encoder layers, thus:

$$\mathbf{W}_{noisy} = \mathbf{W}_{original} + \epsilon_d \quad (5-2)$$

1695 where $\mathbf{W}_{original}$ represents the original weights of the last encoder layers, ϵ_d follows
 1696 a Gaussian distribution with zero mean and variance σ^2 . The variance σ^2 controls
 1697 the level of noise introduced, and thus the degree of disturbance in the resulting
 1698 segmentation masks.

With the application of the encoder layer weight perturbation technique, the resulting noisy masks are shown in Figure 5-5. It can be seen that the morphological consistency is preserved, even though the resulting masks are far from the ground truth annotations, which goes perfectly well for testing the robustness of the models against noisy annotations in crowdsourcing-like scenarios.

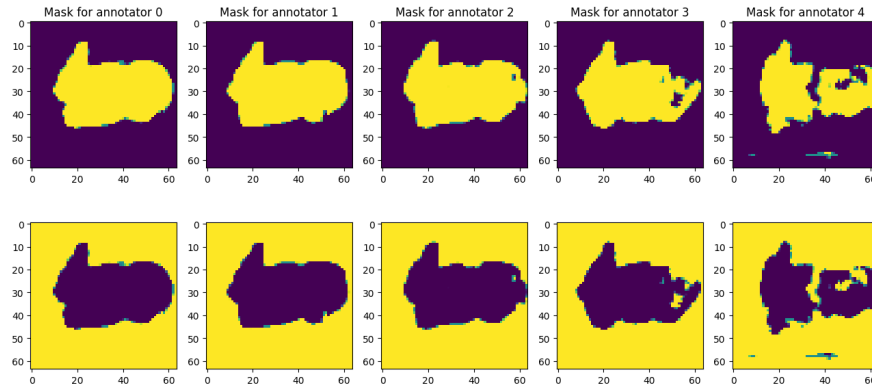


Figure 5-5 Noisy mask generated by enhancing the disturbances in the encoder layers weights for the Oxford-IIIT Pet Dataset. Morphological consistency is preserved. From left to right, SNR levels of noise in the encoder layer are 10, 5, 2, 0, -5 dB. Not adding noisy to the encoder layer would reach to an annotator behavior as good as the originally trained base model.

5.1.2 Missing Labels Emulation

It has also been determined that most real world crowdsourced datasets suffer from missing labels, since not all annotators are able to label all of the input samples. To emulate this behaviour, a missing labels dataset was created by randomly skipping labels from the annotators labels based on a labeling rate $r_l \in [0, 1]$. The labeling rate r_l controls the proportion of labels that are skipped from the annotators labels. The lower the labeling rate, the more labels are skipped from the annotators labels. The implementation of this missing labels emulation together with the noisy annotators generation can be flexibilized through an algorithm, which enables it to be used in

1714 experiments with variable amount of labelers, noise levels, and labeling rates. An
 1715 overview of the algorithm is shown in Algorithm 5-1. A sample of the generated
 1716 noisy and missing labels dataset is shown in Figure 5-6.

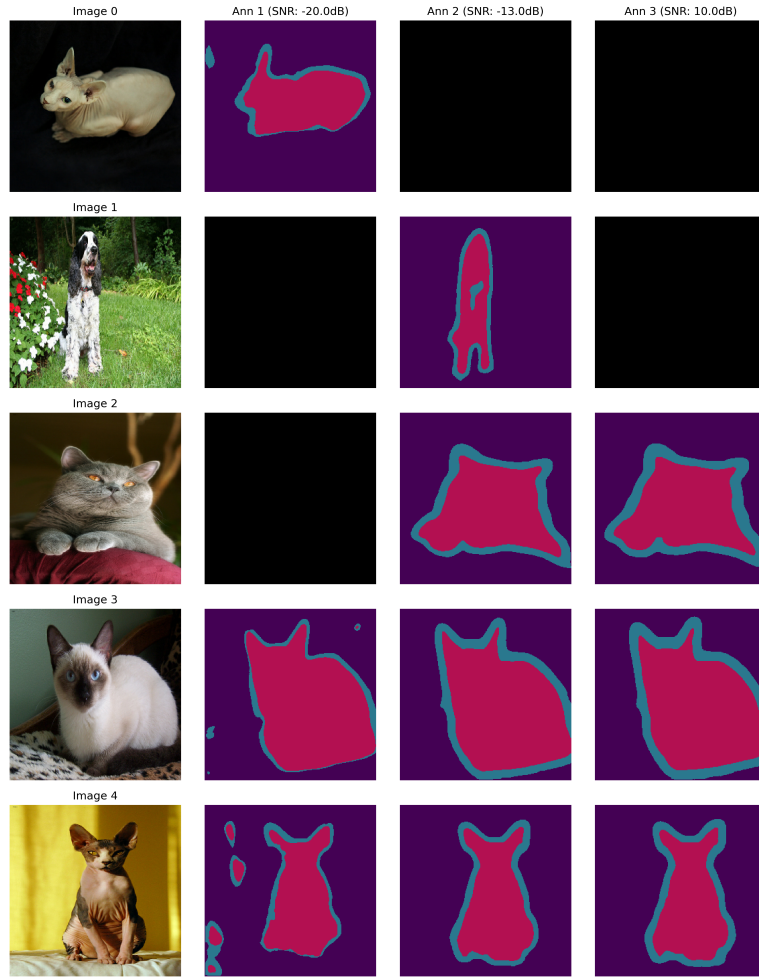


Figure 5-6 Example of missing labels and noisy annotations emulation for the Oxford-IIIT Pet Dataset. The same label rate ($r_l = 0.7$) is preserved for all labelers, which means in this case that every labeler is expected to label 70% of the samples, and skip the remaining 30%.

Algorithm 5-1 Multi-Annotator Dataset Generation with Noisy and Missing Labels

Require: Base segmentation model M , SNR levels $\{s_1, \dots, s_n\}$, labeling rate $r \in [0, 1]$

Ensure: Dataset with multiple noisy annotations and missing labels

```

1: Step 1: Generate Noisy Annotators
2: for each SNR level  $s_i$  do
3:   Create disturbed model  $M_i$  by adding noise to  $M$  with SNR  $s_i$ 
4: end for
5: Step 2: Initialize Labeler Assignment System
6: Create assignment matrix  $A \in \{0, 1\}^{N \times n}$  where  $N$  is number of samples
7: for each annotator  $j \in \{1, \dots, n\}$  do
8:   for each sample  $i \in \{1, \dots, N\}$  do
9:      $A_{i,j} \leftarrow \begin{cases} 1 & \text{with probability } r \\ 0 & \text{with probability } 1 - r \end{cases}$ 
10:   end for
11: end for
12: Step 3: Generate Multi-Annotator Dataset
13: for each image  $x$  in dataset do
14:   Resize  $x$  to target dimensions
15:   for each annotator model  $M_i$  do
16:     Generate annotation  $y_i = M_i(x)$ 
17:     Apply labeler assignment mask:  $y_i \leftarrow y_i \cdot A_{i,j}$ 
18:   end for
19:   Store  $(x, \{y_1, \dots, y_n\})$  in dataset
20: end for
   return Multi-annotator dataset with noisy and missing labels

```

5.2 Results

5.2.1 Oxford-IIIT Pet Dataset

The experimental results on the Oxford-IIIT Pet Dataset demonstrate interesting insights into the performance of different approaches. While STAPLE achieves the highest Dice and Jaccard scores (0.8718 and 0.7917 respectively) for the dataset with no missing labels (label rate of 1.0), our proposed $TGCE_{SS}$ approaches show superior performance for label rates below 1.0 (Table 5-1). The pixel-wise variant, while slightly lower in performance metrics, provides more detailed spatial reliability information. It is of note that the key strength of $TGCE_{SS}$ lies in its ability to simultaneously learn both segmentation and reliability estimation, providing valuable insights into annotator behavior and confidence levels that STAPLE and MV cannot offer.

Method	Label Rate 1.0		Label Rate 0.7		Label Rate 0.3	
	Dice	Jaccard	Dice	Jaccard	Dice	Jaccard
TGCE scalar	0.8622±0.009	0.7791±0.012	0.8568±0.017	0.7726±0.024	0.8529±0.004	0.7677±0.007
TGCE feature	0.6995±0.044	0.5733±0.047	0.4766±0.026	0.3441±0.027	0.4766±0.026	0.3441±0.027
TGCE pixel	0.8468±0.018	0.7589±0.025	0.8545±0.013	0.7695±0.017	0.8283±0.001	0.7359±0.002
MV	0.8132±0	0.7222±0	0.7572±0	0.6554±0	0.5566±0	0.4687±0
STAPLE	0.8718±0	0.7917±0	0.8439±0	0.7609±0	0.5734±0	0.5158±0

Table 5-1 Performance metrics (Dice and Jaccard) at different labeling rates. Best values per category are bolded.

The reliability estimation capabilities of our model are particularly noteworthy, as demonstrated in Figures 5-7 and 5-8. The model successfully captures the varying reliability levels of different annotators, with higher reliability assigned to annotators with lower noise levels (10 dB) and lower reliability to those with higher noise levels (-20 dB). This ability to quantify and visualize annotator reliability is a significant advantage over traditional approaches like STAPLE and MV, which focus solely on segmentation accuracy without providing insights into annotator behavior.

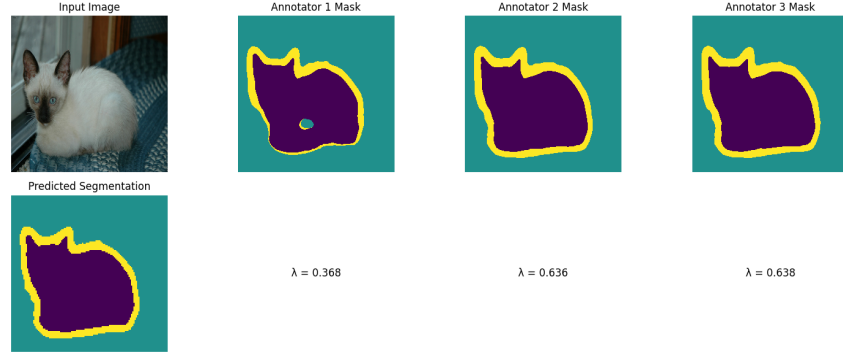


Figure 5-7 Example evaluation of the proposed model in its last trained epoch. An accurate segmentation output is delivered as well as scalar reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0011, $q = 0.4$, noise tolerance = 0.2, $\alpha_{reg} = 0.26$, $\alpha_{entropy} = 0.03$, $\alpha_{sum} = 0.27$.

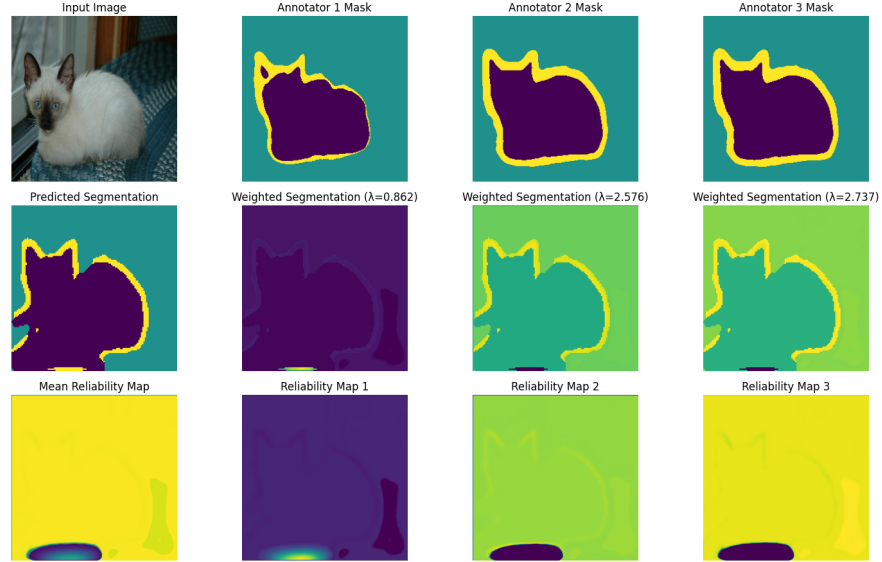


Figure 5-8 Example evaluation of the proposed model in some given epoch (5/50) of training, which demonstrates a quick convergence of the model for correctly estimating the reliability of the labelers masks. An accurate segmentation output is delivered as well as pixel-wise reliability estimations from the labelers masks. Disturbance signal-to-noise ratios are: $[-20, 0, 10]$ dB from left to right. Best found hyperparameters: learning rate = 0.0027, $q = 0.9$, noise tolerance = 0.2, $\alpha_{reg} = 0.38$, $\alpha_{entropy} = 0.02$, $\alpha_{sum} = 0.42$.

5.2.2 TNBC Dataset

As mentioned in Section 2.4.2, [Amgad et al., 2019] presented a Triple Negative Breast Cancer (TNBC) dataset, which is crowdsourced and available as HistomicsTK metadata. Later, [López-Pérez et al., 2023] sampled 151 WSIs, from which 161 ROIs were extracted and labeled by 20 medical students. The dataset was sampled as patches of size 512x512, resulting in 10173 training patches, 1264 validation patches and 399 testing patches. The segmentation masks for the patches are composed of the classes: tumor, stroma, inflammation, necrosis, and others². This subset of the TNBC is named Crowd-Seg.

Raw png files for the images patches and masks were provided by the authors, in practice, in order to optimize the use of these data, a custom tensorflow dataset was built for facilitating batch memory, caching and prefetching. The custom dataset implementation handles the loading and preprocessing of both images and masks, including normalization of images to the [0,1] range and one-hot encoding of the segmentation masks. It also manages the multi-annotator structure by creating a labeler mask that indicates which annotators provided labels for each sample. The dataset supports data augmentation through random transformations (flips, rotations, brightness/contrast adjustments) and includes visualization utilities for debugging and analysis purposes. This custom dataset implementation is publicly available as a PyPI package (see Section 6.3).

Notes on the TNBC Dataset

The TNBC dataset contains a couple of peculiarities that are worth mentioning. First, the label rate (inverse percentage of labels that are missing) is not uniform across the labelers and, in general, quite low, ranging from 1 to 12% as seen in Figure 5-9.

²In practice, some of this masks also contain an additional class labeled as ‘ignore’.

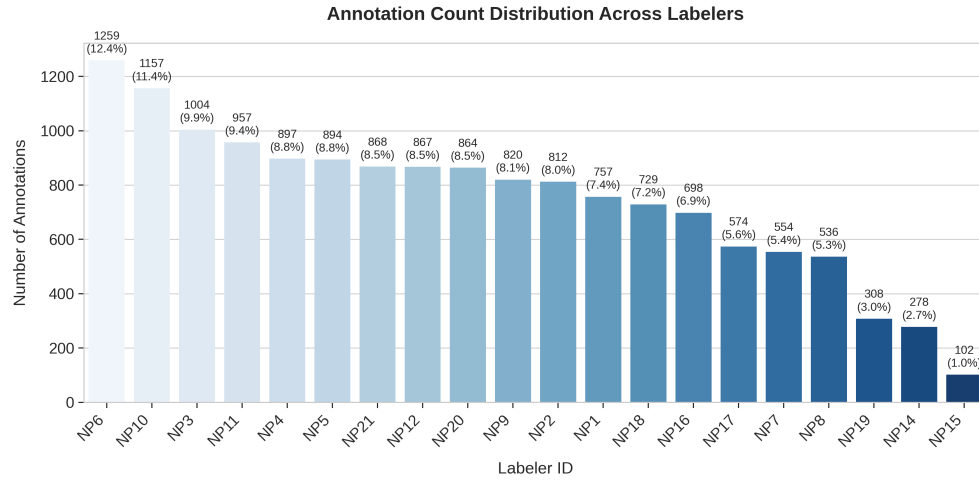


Figure 5-9 Label rate distribution across the labelers for the TNBC (Crowd-Seg) dataset.

On the other hand, the classes distribution across all patches are not equal, being ‘Tumor’ (class 2) the most frequent class. Despite the classes not being equally distributed, the distribution of the classes across the labelers is fairly uniform, as shown in Figure 5-10.

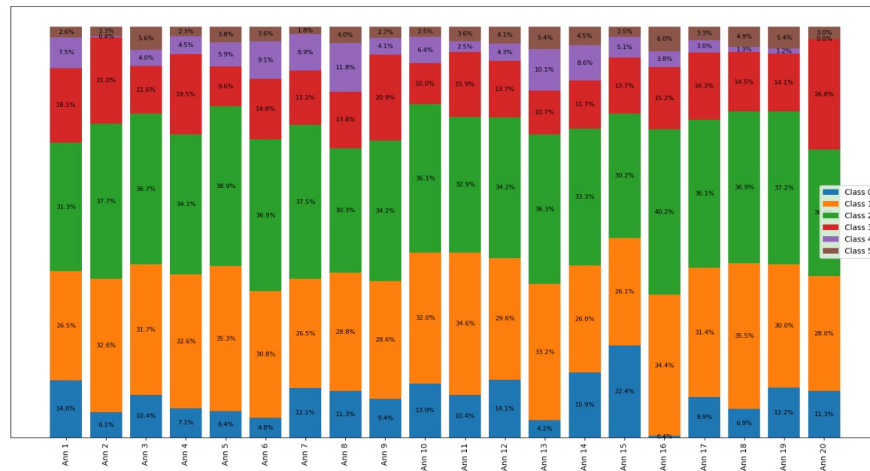


Figure 5-10 Classes distribution across the labelers for the TNBC (Crowd-Seg) dataset. Most common classes are ‘Tumor’ (class 2), ‘Other’ (class 1) and ‘Stroma’ (class 3).

It was also found that the dice score of the labelers masks against the ground truth differed across the labelers and the classes, as shown in Figure 5-11. The target

class (Tumor) ranged dice scores across the labelers from 0.5 to 0.9, other classes
can get as low as 0.4 for in average for some labelers, meaning very high levels of
disagreement.

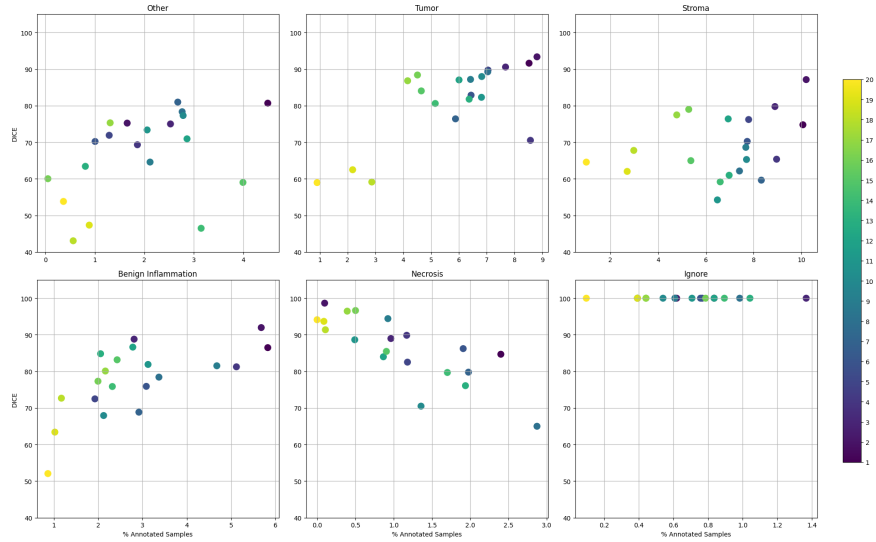


Figure 5-11 Dice score of the labelers masks against the ground truth for the TNBC (Crowd-Seg) dataset.

Finally, it was found that the expert labels, which would thereby be considered as
ground truth for metrics evaluation, were not always correct, as shown in Figure
5-12. These morphological inconsistencies make an objective metric for evaluating
the performance of the proposed model difficult to define, since the ground truth is
not always correct or expressive enough.

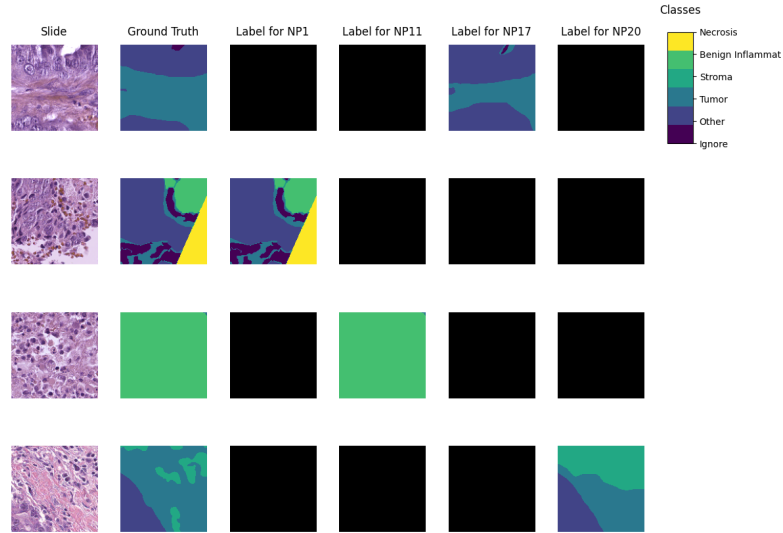


Figure 5-12 Example of ground truth with odd geometries (second patch from top to bottom), which are not always correct or compatible with the underlying structures of the patch itself.

Special considerations for the TNBC Dataset

Because of the aforementioned peculiarities of the TNBC dataset, some special considerations were taken into account for the experiments. Because of the low label rates, experiments showed an unavoidable convergence to estimate arbitrarily high reliabilities on labelers which did not label a certain patch, which is not a desired behavior. None particular element of the proposed loss function penalizes this optimization shortcut, so to avoid this, an additional regularizer was included for penalizing high estimated reliabilities on unlabeled data, as follows:

$$\mathcal{L}_\lambda = \max \left(0, \sum_{\forall i \in R} \frac{1}{1 - \alpha} (\Lambda_i - \mathbb{L}\mathbb{M}_i) \cdot e^{\frac{\Lambda_i - \mathbb{L}\mathbb{M}_i - 1}{\beta}} \right) \quad (5-3)$$

where R is the amount of labelers, Λ_i is the estimated reliability of the labeler i for a certain patch and $\mathbb{L}\mathbb{M}_i$ is the one hot label mask of the labeler i for the same patch. The α and β parameters control the penalty strength for high and low estimated reliabilities, respectively. In the Crowd-Seg experiments, this regularizer replaced the other regularizers, since it was showing better convergence on estimating the reliability of the labelers.



Figure 5-13 Training curves for the baseline model on the TNBC (Crowd-Seg) dataset.

5.3 Results

Initially, a baseline model (ground truth only) was trained with an early stopping mechanism, which stopped the training when the validation loss stopped decreasing for 10 epochs, reaching only 20 epochs of significant training. The results are shown in Figure 5-14.

Then, using the scalar model described in Section 5.0.3, training was performed for 20 epochs and Bayesian hyperparameter optimization on learning rate, decoder dropout rate, α and β for the regularizer, and q and noise tolerance from GCE. The results are shown in Figure 5-15.

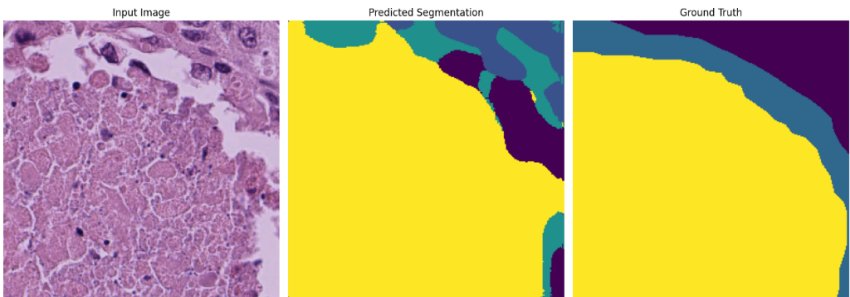


Figure 5-14 Segmentation results for baseline model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Even though the segmentation algorithm seems to show progress on its way to capture the underlying target structures ot be segmented, since the ground truth annotation is not expressive enough, the Dice coefficient remains below 0.55, which is not considered a good performance. The baseline model remained with overfitting even after combining techniques of: Bayesian hyperparameter optimization, data augmentation, learning rate scheduling, dataset shuffling, and early stopping.

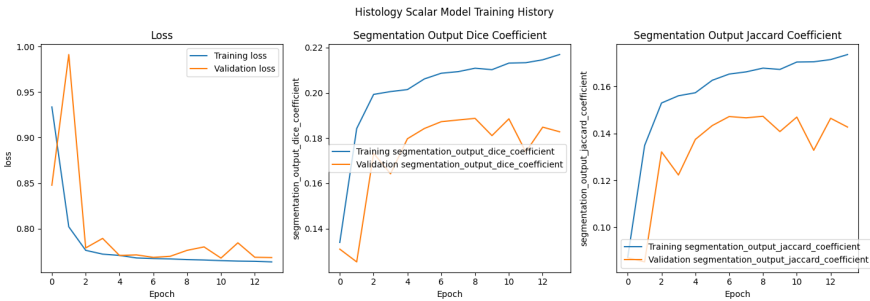


Figure 5-15 Segmentation results for scalar model on the TNBC (Crowd-Seg) dataset after 20 epochs of training. Again, training metrics show a constant grow, but not validation metrics, which shows a combined issue of overfitting and unreliable ground truth.

5.4 Conclusion

The experimental results on both the Oxford-IIIT Pet and TNBC (Crowd-Seg) datasets provide valuable insights into the performance and limitations of our proposed approach. The model demonstrated exceptional performance on the Oxford-IIIT Pet dataset, achieving competitive Dice and Jaccard scores while

successfully capturing annotator reliability patterns. The model’s ability to simultaneously learn segmentation and reliability estimation proved particularly valuable, providing insights into annotator behavior that traditional approaches like STAPLE and MV cannot offer.

However, the experiments on the TNBC dataset revealed significant challenges in evaluating semantic segmentation models for histopathological images when dealing with unreliable ground truth. The morphological inconsistencies in the expert annotations, combined with the extremely low label rates (1-12%) across annotators, made it difficult to objectively measure the model’s performance. While the model showed promising convergence in training, the validation metrics remained unstable, suggesting that the current dataset may not be suitable for rigorous evaluation of multi-annotator segmentation approaches.

These findings highlight the need for more reliable ground truth annotations in histopathological image segmentation datasets. Future work should focus on:

- Developing datasets with more consistent and reliable expert annotations
- Implementing stricter quality control measures for ground truth generation
- Exploring alternative evaluation metrics that account for annotation uncertainty
- Investigating methods to handle extremely sparse annotations more effectively

Despite these challenges, the model’s performance on the Oxford-IIIT Pet dataset demonstrates its effectiveness in handling multiple annotators with varying reliability levels. The model successfully learned to:

- Balance the contributions of different annotators based on their reliability
- Provide detailed spatial reliability information through pixel-wise estimation
- Maintain high segmentation accuracy even with missing labels

- Adapt to different noise levels in annotator inputs

These capabilities make the proposed approach particularly valuable for real-world applications where multiple annotators with varying expertise levels contribute to the dataset. The model's ability to simultaneously learn segmentation and reliability estimation provides a more comprehensive understanding of the annotation process, which could be leveraged to improve the quality of future annotations and dataset construction.

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CHAPTER

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SIX

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FINAL REMARKS

1842

6.1 Conclusions

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This thesis has presented a novel approach to address the challenges of Image Semantic Segmentation (ISS) in histopathology images, particularly in crowdsourcing scenarios where multiple annotators with varying levels of expertise contribute to the labeling process. The proposed solution successfully tackles the fundamental challenges of modeling annotator performance and handling label inconsistencies while maintaining computational efficiency. The work demonstrates significant improvements over existing state-of-the-art methods, particularly in scenarios with noisy and missing labels, which are common in crowdsourcing approaches.

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Throughout this work, several techniques have been explored to develop a model that can face ISS of real world histopathology images in crowdsourcing scenarios. Different approaches were studied and evaluated to reach an optimal solution which could model the annotators' performance from the input space. The following remarks are based on the results obtained from the experiments conducted and reviewed literature:

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- 1858 • Chained Gaussian Processes approaches, while robust for classification and
1859 regression tasks, proved computationally intensive for image segmentation.
1860 This limitation highlighted the need for more efficient approaches that could
1861 handle the complexity of medical image segmentation while modeling
1862 annotator performance.
- 1863 • The development of a novel loss function was crucial for the success of this
1864 work, as existing loss functions were insufficient to model the annotators'
1865 performance across the input space. The proposed loss function successfully
1866 enabled the model to adaptively infer the best possible segmentation without
1867 requiring separate inputs about labeler performance.
- 1868 • The introduction of a tensor map to codify labeler reliability proved to be an
1869 effective approach, allowing the model to implicitly weigh the labelers based
1870 on their performance across the mask and classes space, thus handling the
1871 labeler performance inconsistency effectively.
- 1872 • The use of U-Shaped Deep Learning models as a building block for the
1873 proposed solution was validated, as it effectively captured spatial
1874 relationships between pixels and annotations while handling the variability
1875 in image quality.
- 1876 • The combination of Chained Deep Learning approaches with the proposed
1877 loss function and tensor map representation resulted in a comprehensive
1878 solution being, to the best of our knowledge, the first approach on offering
1879 an image segmentation solution for multiple labelers in medical image with
1880 interpretability in the reliability of the labelers, successfully addressing both
1881 labeler inconsistency and image quality variability. The model outperformed
1882 other state of the art solutions in terms of segmentation accuracy in noisy
1883 and missing labels conditions, which is rather common in crowdsourcing
1884 approaches.

6.2 Future work

As tools and frameworks for bayesian approaches like Chained Gaussian Processes¹ evolve, it is expected to see a decrease in computational requirements and hence, making mixed models more appealing for the task of ISS of histopathology images. In that sense, the use of CCGPMA as a building block for the proposed solution could be a good approach, as it is a powerful model that can take advantage of the inter-dependencies between the annotators, and the powerfulness of CNNs to extract features from the images.

The noisy labelers emulation technique developed in this work presents an opportunity to test the model's robustness in various medical image segmentation challenges. This technique could be adapted to simulate different types of expert variability in other medical imaging domains:

- **Multi-modal Medical Imaging:** The technique could be extended to simulate expert disagreements in multi-modal scenarios, where different imaging modalities (e.g., MRI, CT, PET) might lead to varying interpretations of the same anatomical structure.
- **Temporal Consistency:** For longitudinal studies, the emulation could be modified to simulate how expert annotations might vary over time, helping to test the model's ability to maintain consistency in segmentation across different time points.
- **Cross-institutional Variability:** The noise injection technique could be calibrated to simulate the variations in annotation patterns that arise from different institutional protocols or training backgrounds.

¹Including GPFlow, and GPFlux.

1908 • **Pathology-specific Challenges:** The emulation could be tailored to simulate
1909 expert disagreements in specific pathological conditions, where the
1910 morphological features might be more ambiguous or subject to
1911 interpretation.

1912 These extensions would help validate the model's performance across a broader
1913 range of medical imaging scenarios and ensure its robustness in real-world clinical
1914 applications.

1915 The pixel-wise reliability mapping approach shows particular promise for future
1916 development. While the current implementation already demonstrates good
1917 performance, several enhancements could lead to significant improvements:

1918 • **Advanced Data Augmentation:** Implementing a comprehensive data
1919 augmentation strategy could significantly improve the model's convergence
1920 and performance. This could include:

- 1921 – Geometric transformations (rotations, flips, elastic deformations)
- 1922 – Intensity variations to simulate different staining conditions
- 1923 – Multi-scale training to handle varying tissue sizes
- 1924 – Adversarial training to improve robustness

1925 These augmentations would help the model learn more robust features and
1926 better generalize to unseen data.

1927 • **Attention Mechanisms:** Integrating attention mechanisms into the pixel-wise
1928 reliability branch could help the model focus on the most relevant regions for
1929 reliability estimation, potentially improving both accuracy and computational
1930 efficiency.

1931 • **Multi-scale Reliability Estimation:** Implementing a hierarchical approach to
1932 reliability estimation could capture both fine-grained local variations and
1933 broader regional patterns in annotator reliability, this would imply a
1934 network with yet more branches and outputs.

The combination of these enhancements with the existing pixel-wise reliability mapping approach could potentially outperform all other techniques, as it would provide the most detailed and accurate representation of annotator reliability while maintaining computational efficiency. The use of data augmentation, in particular, could help the model converge faster and achieve better generalization, as it would expose the model to a wider variety of possible variations in both the input images and the corresponding annotations.

6.3 Academic Discussion

To facilitate the adoption and further development of the proposed approach, several resources have been made publicly available. A comprehensive GitHub repository has been created that provides easy-to-use APIs for implementing the proposed solution. The repository includes pre-built dataset loaders for TensorFlow/Keras frameworks, implementation of the $TGCE_{SS}$ loss function in all its variants (scalar, features, and pixel-wise), and a neural network architecture builder with experimental examples. A key feature of this repository is its automatic dataset fetching from AWS S3, which eliminates the need for manual downloads or external dependencies. This repository aims to lower the entry barrier for researchers and practitioners interested in implementing the proposed approach.

To further simplify the adoption of the proposed solution, a PyPI package has been published with detailed documentation available at <https://seg-tgce.readthedocs.io/en/latest/>. The documentation provides comprehensive installation instructions, API reference, usage examples, and best practices for implementation. This package makes it straightforward for the community to integrate the proposed approach into their existing workflows, with a focus on ease of use and clear documentation of all available features and options.

1962 An academic publication is currently being prepared for submission to a Q1
1963 journal. This publication will provide detailed theoretical foundations of the
1964 proposed approach, comprehensive experimental results and analysis, comparison
1965 with state-of-the-art methods, and discussion of limitations and future directions.
1966 The publication will serve as the primary reference for the theoretical and practical
1967 aspects of the proposed solution, ensuring that the work is properly documented
1968 and validated through the peer-review process.

1969 These resources collectively contribute to the reproducibility and accessibility of
1970 the research, enabling the broader scientific community to build upon and extend
1971 the proposed approach. The combination of open-source code, well-documented
1972 APIs, and peer-reviewed publication ensures that the work can be properly
1973 evaluated, validated, and utilized by other researchers in the field. This
1974 comprehensive approach to sharing research outputs aligns with current best
1975 practices in scientific computing and machine learning research, promoting both
1976 transparency and practical utility of the proposed solution.

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